VC02_VC03

April 12, 2026

0.1 NUESTRA PRIMERA RED NEURONAL: MNIST DATASET

[1]:
```
!pip install $packageName
```

```
ERROR: Invalid requirement: '$packageName': Expected package name at the start
of dependency specifier
    $packageName
    ^
```

[2]:
```
!pip freeze
```

```
absl-py==2.4.0
aiobotocore @ file:///C:/miniconda3/conda-bld/aiobotocore_1760994335546/work
aiodns @ file:///C:/miniconda3/conda-bld/aiodns_1762185075564/work
aiohappyeyeballs @
file:///C:/b/abs_f806x82huz/croot/aiohappyeyeballs_1755769855628/work
aiohttp @ file:///C:/miniconda3/conda-bld/aiohttp_1762519774760/work
aioitertools @ file:///C:/b/abs_70vydn6fmg/croot/aioitertools_1754313763510/work
aiosignal @ file:///C:/b/abs_84ej696pud/croot/aiosignal_1755870823775/work
alabaster @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/alabaster_1729041938345/work
altair @ file:///C:/b/abs_0b1avjyzoc/croot/altair_1743016738596/work
anaconda-anon-usage @ file:///opt/miniconda3/conda-bld/anaconda-anon-
usage_1764636648062/work
anaconda-auth @ file:///C:/miniconda3/conda-bld/anaconda-cloud-auth-
split_1765839853030/work
anaconda-catalogs @ file:///C:/miniconda3/conda-bld/anaconda-
catalogs_1764859031645/work
anaconda-cli-base @ file:///C:/miniconda3/conda-bld/anaconda-cli-
base_1764888883727/work
anaconda-client @ file:///C:/miniconda3/conda-bld/anaconda-
client_1763659439838/work
anaconda-navigator @ file:///C:/miniconda3/conda-bld/anaconda-
navigator_1761578688358/work
anaconda-project @ file:///C:/b/abs_c3nd60wjs2/croot/anaconda-
project_1746215311215/work
annotated-types==0.7.0
anyio==3.7.1
appdirs==1.4.4
```

```
archspec @ file:///home/task_175812491784513/conda-bld/archspec_1758124989039/work
argon2-cffi @ file:///opt/conda/conda-bld/argon2-cffi_1645000214183/work
argon2-cffi-bindings @ file:///C:/miniconda3/conda-bld/argon2-cffi-bindings_1757925023673/work
arrow @ file:///C:/miniconda3/conda-bld/arrow_1761560549461/work
astroid @ file:///C:/miniconda3/conda-bld/astroid_1759355341836/work
astropy @ file:///C:/miniconda3/conda-bld/astropy_1762433878873/work
astropy-iers-data @ file:///C:/miniconda3/conda-bld/astropy-iers-data_1763019997878/work
asttokens @ file:///C:/b/abs_9662ywy9fp/croot/asttokens_1743630464377/work
astunparse==1.6.3
async-lru @ file:///C:/miniconda3/conda-bld/async-lru_1761121577604/work
asyncssh @ file:///C:/miniconda3/conda-bld/asyncssh_1763120359147/work
atomicwrites==1.4.0
attrs @ file:///C:/miniconda3/conda-bld/attrs_1762356899359/work
Automat @ file:///C:/b/abs_e6j20ekmp5/croot/automat_1743532580167/work
autopep8 @ file:///croot/autopep8_1708962882016/work
babel @ file:///C:/miniconda3/conda-bld/babel_1764159345001/work
bcrypt @ file:///C:/miniconda3/conda-bld/bcrypt_1762875213278/work
beautifulsoup4==4.14.3
binaryornot @ file:///tmp/build/80754af9/binaryornot_1617751525010/work
black @ file:///C:/miniconda3/conda-bld/black_1760632469648/work
bleach @ file:///C:/miniconda3/conda-bld/bleach_1764153675285/work
blinker @ file:///C:/b/abs_b1i87khtob/croot/blinker_1737448732095/work
bokeh @ file:///C:/miniconda3/conda-bld/bokeh_1757102818337/work
boltons @ file:///C:/b/abs_e2_iokhxbp/croot/boltons_1751383740243/work
botocore @ file:///C:/miniconda3/conda-bld/botocore_1760965317697/work
Bottleneck @ file:///C:/miniconda3/conda-bld/bottleneck_1761938102004/work
brotlicffi @ file:///C:/miniconda3/conda-bld/brotlicffi_1762950508507/work
cachetools @ file:///C:/b/abs_fcvwpc5vh9/croot/cachetools_1738224687938/work
cattrs @ file:///C:/miniconda3/conda-bld/cattrs_1761729120648/work
certifi==2026.1.4
cffi @ file:///C:/miniconda3/conda-bld/cffi_1761832792955/work
chardet @ file:///C:/b/abs_3f08pnpj_d/croot/chardet_1755516293556/work
charset-normalizer==3.4.4
click==8.3.1
cloudpickle @ file:///C:/b/abs_fdydcrc9_2/croot/cloudpickle_1751309451995/work
colorama==0.4.6
colorcet @ file:///C:/Users/dev-admin/perseverance-python-buildout/croot/colorcet_1729053366498/work
comm @ file:///C:/miniconda3/conda-bld/comm_1763119281973/work
conda @ file:///C:/miniconda3/conda-bld/conda_1765567840841/work/conda-src
conda-anaconda-telemetry @ file:///croot/conda-anaconda-telemetry_1755883788794/work
conda-anaconda-tos @ file:///C:/b/abs_c9yejtsx9g/croot/conda-anaconda-tos_1755123332641/work
conda-build @ file:///C:/miniconda3/conda-bld/conda-build_1764886947283/work
```

```
conda-content-trust @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/conda-content-trust_1729088072778/work
conda-libmamba-solver @ file:///C:/miniconda3/conda-bld/conda-libmamba-
solver_1764245615235/work/src
conda-pack @ file:///C:/b/abs_7b5jwr84ex/croot/conda-pack_1752674284442/work
conda-package-handling @ file:///C:/miniconda3/conda-bld/conda-package-
handling_1762366515145/work
conda-repo-cli @ file:///C:/b/abs_e2mlfkr6jb/croot/conda-repo-
cli_1755793053578/work
conda_index @ file:///C:/miniconda3/conda-bld/conda-index_1761321711844/work
conda_package_streaming @ file:///C:/miniconda3/conda-bld/conda-package-
streaming_1762361689679/work
constantly @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/constantly_1729053406505/work
contourpy @ file:///C:/miniconda3/conda-bld/contourpy_1763643940481/work
cookiecutter @ file:///C:/b/abs_6ak0be77em/croot/cookiecutter_1751313989831/work
cryptography @ file:///C:/miniconda3/conda-bld/cryptography-
split_1761932023265/work
cssselect @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/cssselect_1729047831562/work
cycler @ file:///tmp/build/80754af9/cycler_1637851556182/work
dacite==1.9.2
dask @ file:///C:/miniconda3/conda-bld/dask-core_1763019062292/work
datashader @ file:///C:/b/abs_0dvz2ks6tk/croot/datashader_1756382478739/work
debugpy @ file:///C:/miniconda3/conda-bld/debugpy_1762421658007/work
decorator @ file:///C:/miniconda3/conda-bld/decorator_1757341249916/work
defusedxml @ file:///tmp/build/80754af9/defusedxml_1615228127516/work
Deprecated==1.3.1
diff-match-patch @ file:///Users/ktietz/demo/mc3/conda-bld/diff-match-
patch_1630511840874/work
dill @ file:///C:/miniconda3/conda-bld/dill_1760096877594/work
distributed @ file:///C:/miniconda3/conda-bld/distributed_1763105975646/work
distro @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/distro_1729059153117/work
docstring-to-markdown @ file:///C:/miniconda3/conda-bld/docstring-to-
markdown_1758626863507/work
docutils @ file:///C:/miniconda3/conda-bld/docutils_1761746640431/work
et_xmlfile @ file:///C:/b/abs_b0ifp_j3js/croot/et_xmlfile_1756462499666/work
evalidate @ file:///C:/b/abs_573nc1ymt6/croot/evalidate_1743606028415/work
executing @ file:///C:/miniconda3/conda-bld/executing_1757061255268/work
fastapi==0.103.2
fastjsonschema @ file:///C:/miniconda3/conda-bld/python-
fastjsonschema_1763718729498/work
filelock @ file:///C:/miniconda3/conda-bld/filelock_1762419942375/work
flake8 @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/flake8_1729054074515/work
Flask @ file:///C:/b/abs_28dbmjidmh/croot/flask_1756453142071/work
flatbuffers==25.12.19
```

```
fonttools @ file:///C:/miniconda3/conda-bld/fonttools_1759754890124/work
frozendict @ file:///C:/miniconda3/conda-bld/frozendict_1761750728192/work
frozenlist @ file:///C:/miniconda3/conda-bld/frozenlist_1761062034652/work
fsspec @ file:///C:/miniconda3/conda-bld/fsspec_1762432453760/work
gast==0.7.0
gazpacho==1.1
gitdb @ file:///C:/b/abs_1bkmriimss/croot/gitdb_1753362133190/work
GitPython @ file:///C:/miniconda3/conda-bld/gitpython_1761291338863/work
gmpy2 @ file:///C:/b/abs_d8ki0o0h97/croot/gmpy2_1738085498525/work
google-pasta==0.2.0
greenlet @ file:///C:/miniconda3/conda-bld/greenlet_1757405549674/work
grpcio==1.80.0
h11==0.16.0
h5py==3.14.0
HeapDict @ file:///Users/ktietz/demo/mc3/conda-bld/heapdict_1630598515714/work
holoviews @ file:///C:/miniconda3/conda-bld/holoviews_1763365342364/work
html5lib @ file:///Users/ktietz/demo/mc3/conda-bld/html5lib_1629144453894/work
httpcore @ file:///C:/b/abs_25m7_xthp6/croot/httpcore_1748526065845/work
httpx @ file:///C:/miniconda3/conda-bld/httpx_1760447475275/work
hvplot @ file:///C:/miniconda3/conda-bld/hvplot_1756992721648/work
hyperlink @ file:///tmp/build/80754af9/hyperlink_1610130746837/work
idna==3.11
ImageIO @ file:///C:/miniconda3/conda-bld/imageio_1762954225001/work
imagesize @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/imagesize_1729042613392/work
imbalanced-learn @ file:///C:/b/abs_d1hr2znu7o/croot/imbalanced-
learn_1756460314059/work
img2pdf==0.6.3
importlib_metadata @ file:///C:/miniconda3/conda-bld/importlib_metadata-
suite_1763996638226/work
incremental @ file:///croot/incremental_1743512672142/work
inflection @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/inflection_1729054658204/work
iniconfig @ file:///C:/b/abs_70kaor_ccd/croot/iniconfig_1752653749904/work
intake @ file:///C:/b/abs_36v90t4ek6/croot/intake_1752749551025/work
intervaltree @ file:///Users/ktietz/demo/mc3/conda-
bld/intervaltree_1630511889664/work
ipykernel @ file:///C:/miniconda3/conda-bld/ipykernel_1762946092426/work
ipympl @ file:///C:/b/abs_2fdfe_fnms/croot/ipympl_1752060862279/work
ipython @ file:///C:/miniconda3/conda-bld/ipython_1762789322374/work
ipython_pygments_lexers @
file:///C:/b/abs_b66hj1lo19/croot/ipython_pygments_lexers_1744753262043/work
ipywidgets @ file:///C:/miniconda3/conda-bld/ipywidgets_1761560177683/work
isort @ file:///C:/miniconda3/conda-bld/isort_1760089000590/work
itemadapter @ file:///C:/b/abs_87vsk6a5ji/croot/itemadapter_1756732900203/work
itemloaders @ file:///C:/b/abs_f35dqz89dk/croot/itemloaders_1743513855167/work
itsdangerous @ file:///C:/Users/dev-
admin/buildout/croot/itsdangerous_1732929022589/work
```

```
jaconv==0.4.1
jaraco.classes @
file:///C:/b/abs_6erueoob1v/croot/jaraco.classes_1755516340851/work
jaraco.context @ file:///C:/Users/dev-admin/buildout/perseverance-python-
buildout/croot/jaraco.context_1731721000658/work
jaraco.functools @
file:///C:/b/abs_a1jv6v_pzp/croot/jaraco.functools_1740408129620/work
jedi @ file:///C:/b/abs_3a2kbnlclc/croot/jedi_1733987412687/work
jellyfish @ file:///C:/miniconda3/conda-bld/jellyfish_1764243816637/work
Jinja2 @ file:///C:/b/abs_920kup4e6u/croot/jinja2_1741711580669/work
jmespath @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/jmespath_1729040936778/work
joblib @ file:///C:/miniconda3/conda-bld/joblib_1757926362684/work
json5 @ file:///C:/miniconda3/conda-bld/json5_1763718601351/work
jsonpatch @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/jsonpatch_1729054776004/work
jsonpointer @ file:///C:/b/abs_73u73l7pl9/croot/jsonpointer_1753788460913/work
jsonschema @ file:///C:/miniconda3/conda-bld/jsonschema_1762459308681/work
jsonschema-specifications @ file:///C:/miniconda3/conda-bld/jsonschema-
specifications_1762788613133/work
jupyter @ file:///C:/Users/dev-admin/buildout/croot/jupyter_1742594998199/work
jupyter-console @ file:///C:/Users/dev-
admin/buildout/croot/jupyter_console_1739561496258/work
jupyter-events @
file:///C:/b/abs_9cm3qlticu/croot/jupyter_events_1741184612840/work
jupyter-lsp @ file:///C:/b/abs_7171flzdkg/croot/jupyter-lsp-
meta_1745827033118/work
jupyter_client @ file:///C:/miniconda3/conda-
bld/jupyter_client_1762943100578/work
jupyter_core @ file:///C:/b/abs_7c6fipznw3/croot/jupyter_core_1751991383524/work
jupyter_server @
file:///C:/b/abs_cdpu8und98/croot/jupyter_server_1751996637146/work
jupyter_server_terminals @
file:///C:/b/abs_adjrm9dtns/croot/jupyter_server_terminals_1744706714294/work
jupyterlab @ file:///C:/miniconda3/conda-bld/jupyterlab_1758125082061/work
jupyterlab_pygments @
file:///C:/b/abs_d5alfet8m6/croot/jupyterlab_pygments_1741124274578/work
jupyterlab_server @ file:///C:/miniconda3/conda-
bld/jupyterlab_server_1761751443648/work
jupyterlab_widgets @
file:///C:/b/abs_6566by8dag/croot/jupyterlab_widgets_1752066574111/work
keras==3.14.0
keyring @ file:///C:/miniconda3/conda-bld/keyring_1763637203252/work
kiwisolver @ file:///C:/miniconda3/conda-bld/kiwisolver_1764159357446/work
lazy_loader @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/lazy_loader_1729054947204/work
lckr_jupyterlab_variableinspector @
file:///C:/b/abs_a7k1hndm37/croot/jupyterlab-
```

```
variableinspector_1744638305846/work
libarchive-c @ file:///croot/python-libarchive-c_1726069797193/work
libclang==18.1.1
libmambapy @ file:///C:/miniconda3/conda-bld/mamba-
split_1763111608356/work/libmambapy
lief @ file:///C:/miniconda3/conda-bld/lief_1762942044025/work/api/python
linkify-it-py @ file:///C:/miniconda3/conda-bld/linkify-it-py_1758717669406/work
llvmlite @ file:///C:/miniconda3/conda-bld/llvmlite_1760469323312/work
lmdb @ file:///C:/miniconda3/conda-bld/python-lmdb_1764243825905/work
locket @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/locket_1729042822914/work
lsprotocol @ file:///C:/miniconda3/conda-
bld/lsprotocol_1759828550184/work/packages/python
lxml==6.0.2
lz4 @ file:///C:/miniconda3/conda-bld/lz4_1762954228844/work
Markdown @ file:///C:/b/abs_f9hbi7znfr/croot/markdown_1746114643651/work
markdown-it-py==4.0.0
MarkupSafe @ file:///C:/b/abs_a0ma7ge0jc/croot/markupsafe_1738584052792/work
matplotlib==3.10.6
matplotlib-inline @ file:///C:/miniconda3/conda-bld/matplotlib-
inline_1762779276169/work
mccabe @ file:///opt/conda/conda-bld/mccabe_1644221741721/work
mdit-py-plugins @ file:///C:/b/abs_88j0o6_jpr/croot/mdit-py-
plugins_1756381038807/work
mdurl==0.1.2
menuinst @ file:///C:/miniconda3/conda-bld/menuinst_1765382397455/work
mistune @ file:///C:/b/abs_77yql3poyz/croot/mistune_1741124004410/work
mkl-service==2.5.2
mkl_fft @ file:///C:/miniconda3/conda-bld/mkl_fft_1761592916781/work
mkl_random @ file:///C:/miniconda3/conda-bld/mkl_random_1761593076691/work
ml_dtypes==0.5.4
more-itertools @ file:///C:/miniconda3/conda-bld/more-
itertools_1761121564395/work
mpi4py @ file:///C:/b/abs_2c72fs0pz9/croot/mpi4py_1749458512296/work
mpmath @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/mpmath_1729048934933/work
msgpack @ file:///C:/b/abs_4b3t4uhz3r/croot/msgpack-python_1750958631084/work
multidict @ file:///C:/miniconda3/conda-bld/multidict_1760566895443/work
multipledispatch @ file:///C:/miniconda3/conda-
bld/multipledispatch_1756978485717/work
mutagen==1.47.0
mypy @ file:///C:/miniconda3/conda-bld/mypy-split_1761904873187/work
mypy_extensions @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/mypy_extensions_1729049023435/work
namex==0.1.0
narwhals @ file:///C:/miniconda3/conda-bld/narwhals_1760003513313/work
navigator-updater @ file:///C:/miniconda3/conda-bld/navigator-
updater_1761563279412/work
```

```
nbclient @ file:///C:/b/abs_a09c4t3h8x/croot/nbclient_1741124030330/work
nbconvert @ file:///C:/b/abs_c27_60dzt8/croot/nbconvert-meta_1741191385337/work
nbformat @ file:///C:/Users/dev-admin/buildout/croot/nbformat_1739559077024/work
nest_asyncio @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/nest-asyncio_1729042999160/work
networkx @ file:///C:/b/abs_6c02vx1ct4/croot/networkx_1756709289161/work
nltk @ file:///C:/miniconda3/conda-bld/nltk_1761121193106/work
notebook @ file:///C:/b/abs_ddnsjx06fh/croot/notebook_1756709315423/work
notebook_shim @ file:///C:/b/abs_9ctyfgpncn/croot/notebook-
shim_1741707829491/work
numba @ file:///C:/miniconda3/conda-bld/numba_1760991948802/work
numexpr @ file:///C:/miniconda3/conda-bld/numexpr_1762165724132/work
numpy @ file:///C:/miniconda3/conda-bld/numpy_and_numpy_base_1763980694785/work/
dist/numpy-2.3.5-cp313-cp313-
win_amd64.whl#sha256=242110f8490d8f1e8ed7b1fb286cc718d5e96f1ba89bee9e7f807ccf5cb
e59e6
numpydoc @ file:///C:/b/abs_e63bcy53_z/croot/numpydoc_1750958653347/work
opencv-python==4.13.0.92
openpyxl @ file:///C:/b/abs_6cdczthntz/croot/openpyxl_1736371239628/work
opt_einsum==3.4.0
optree==0.19.0
overrides @ file:///C:/miniconda3/conda-bld/overrides_1762349109990/work
packaging==26.0
pandas @ file:///C:/miniconda3/conda-bld/pandas_1762332401545/work/dist/pandas-
2.3.3-cp313-cp313-
win_amd64.whl#sha256=9d93408ded21463dc4a0cce0d61428cd756d3f130f78ca53bfb6d1c81fe
daa66
pandocfilters @ file:///C:/miniconda3/conda-bld/pandocfilters_1756977623249/work
panel @ file:///C:/miniconda3/conda-bld/panel_1762899230457/work
param @ file:///C:/miniconda3/conda-bld/param_1763638469890/work
parsel @ file:///C:/miniconda3/conda-bld/parsel_1761560684296/work
parso @ file:///C:/miniconda3/conda-bld/parso_1762781952168/work
partd @ file:///C:/b/abs_e1k4ow1dv4/croot/partd_1736803443606/work
pathspec @ file:///C:/miniconda3/conda-bld/pathspec_1761820745063/work
patsy @ file:///C:/b/abs_f9h0eiedk0/croot/patsy_1738159949276/work
pexpect @ file:///C:/miniconda3/conda-bld/pexpect_1762535956813/work
pickleshare @ file:///tmp/build/80754af9/pickleshare_1606932040724/work
pikepdf==10.3.0
pillow==12.1.1
pkce @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/pkce_1729049216383/work
pkginfo @ file:///C:/miniconda3/conda-bld/pkginfo_1765439573823/work
platformdirs==4.5.1
plotly @ file:///C:/b/abs_8926yyyylu/croot/plotly_1756453681452/work
pluggy @ file:///C:/b/abs_dfec_m79vo/croot/pluggy_1733170145382/work
prometheus_client @
file:///C:/b/abs_8b175q_ub8/croot/prometheus_client_1744271638821/work
prompt_toolkit @ file:///C:/miniconda3/conda-bld/prompt-
```

```
toolkit_1761745018781/work
propcache @ file:///C:/b/abs_57780xb0gk/croot/propcache_1744012740137/work
Protego @ file:///C:/b/abs_48sti5uql_/croot/protego_1746479771804/work
protobuf==7.34.1
psutil @ file:///C:/miniconda3/conda-bld/psutil_1761896538278/work
ptyprocess @ file:///opt/miniconda3/conda-
bld/ptyprocess_1762424170819/work/dist/ptyprocess-0.7.0-py2.py3-none-
any.whl#sha256=3470be7f810474c8a2ecfcd6e02acc6aea8483ab595417fa4e336362a349933e
pure_eval @ file:///C:/miniconda3/conda-bld/pure_eval_1757067099531/work
py-cpuinfo @ file:///C:/Users/dev-admin/perseverance-python-buildout/croot/py-
cpuinfo_1729051209925/work
pyarrow @ file:///C:/miniconda3/conda-bld/pyarrow_1759833600190/work/python
pyasn1 @ file:///C:/Users/dev-admin/buildout/perseverance-python-
buildout/croot/pyasn1_1731744867547/work
pyasn1_modules @
file:///C:/b/abs_12mhyss7ds/croot/pyasn1-modules_1752042911584/work
pycares @ file:///C:/b/abs_b36fcwt8zj/croot/pycares_1756217945340/work
pycodestyle @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/pycodestyle_1729051264017/work
pycosat @ file:///C:/b/abs_18nblzzn70/croot/pycosat_1736868434419/work
pycparser @ file:///C:/miniconda3/conda-bld/pycparser_1757496153123/work
pyct @ file:///C:/miniconda3/conda-bld/pyct_1759932166264/work
pycurl @ file:///C:/miniconda3/conda-bld/pycurl_1761132136915/work
pydantic==2.12.5
pydantic-settings @ file:///C:/miniconda3/conda-bld/pydantic-
settings_1764165236385/work
pydantic_core==2.41.5
PyDispatcher @ file:///C:/b/abs_7bay5wbi80/croot/pydispatcher_1750159265590/work
pydocstyle @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/pydocstyle_1729055775306/work
pyerfa @ file:///C:/b/abs_35x4fn08_4/croot/pyerfa_1738083087918/work
pyflakes @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/pyflakes_1729051360949/work
PyGithub @ file:///C:/miniconda3/conda-bld/pygithub_1762271604729/work
Pygments==2.19.2
PyJWT @ file:///C:/Users/dev-admin/buildout/croot/pyjwt_1739559988030/work
pykakasi==2.3.0
pylint @ file:///C:/b/abs_6dheb4r34g/croot/pylint_1756473040471/work
pylint-venv @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/pylint-venv_1729062196487/work
pyls-spyder==0.4.0
PyNaCl @ file:///C:/miniconda3/conda-bld/pynacl_1758892069004/work
pyodbc @ file:///C:/miniconda3/conda-bld/pyodbc_1761670569043/work
pyOpenSSL @ file:///C:/miniconda3/conda-bld/pyopenssl_1762166557109/work
pyparsing @ file:///C:/miniconda3/conda-bld/pyparsing_1763974011379/work
PyQt5==5.15.11
PyQt5_sip @ file:///C:/miniconda3/conda-bld/pyqt-
split_1764009081060/work/pyqt_sip
```

```
PyQtWebEngine==5.15.7
PySide6==6.9.2
PySocks==1.7.1
pytest @ file:///C:/miniconda3/conda-bld/pytest_1758125117201/work
python-dateutil==2.9.0.post0
python-dotenv @ file:///C:/b/abs_71cpoh9hpg/croot/python-
dotenv_1745613639902/work
python-json-logger @ file:///C:/b/abs_0cm_mnox0z/croot/python-json-
logger_1734370042436/work
python-lsp-black @ file:///C:/b/abs_56xq_70uqy/croot/python-lsp-
black_1744023225294/work
python-lsp-jsonrpc @ file:///croot/python-lsp-jsonrpc_1708962872556/work
python-lsp-ruff @ file:///C:/miniconda3/conda-bld/python-lsp-
ruff_1759832197408/work
python-lsp-server @ file:///C:/miniconda3/conda-bld/python-lsp-
server_1759829143457/work
python-slugify==8.0.4
pytokens @ file:///C:/miniconda3/conda-bld/pytokens_1764151940697/work
pytoolconfig @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/pytoolconfig_1729051670797/work
pytube==15.0.0
pytz @ file:///C:/b/abs_f8wdzeix0n/croot/pytz_1752135878094/work
pyuca @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/pyuca_1729099690749/work
pyviz_comms @ file:///C:/miniconda3/conda-bld/pyviz_comms_1758102131608/work
PyWavelets @ file:///C:/b/abs_ebd1bcubeh/croot/pywavelets_1756715672892/work
pywin32 @ file:///C:/b/abs_6c9w_vp9zi/croot/pywin32_1754670534373/work
pywin32-ctypes @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/pywin32-ctypes_1729046491215/work
pywinpty @ file:///C:/b/abs_883wh7sts8/croot/pywinpty_1741871674963/work
PyYAML @ file:///C:/miniconda3/conda-bld/pyyaml_1763465513644/work
pyzmq @ file:///C:/miniconda3/conda-bld/pyzmq_1762375609051/work
QDarkStyle @ file:///croot/qdarkstyle_1709231003551/work
qstylizer @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/qstylizer_1729063143005/work/dist/qstylizer-0.2.2-py2.py3-none-
any.whl#sha256=9e6ed4af5d7d32c75075c807064c85e792b62db8b603152b16608d504e3ed165
QtAwesome @ file:///C:/b/abs_f3usq20nxv/croot/qtawesome_1744810809443/work
qtconsole @ file:///C:/miniconda3/conda-bld/qtconsole_1758748755800/work
QtPy @ file:///C:/miniconda3/conda-bld/qtpy_1757531176201/work
queuelib @ file:///C:/miniconda3/conda-bld/queuelib_1762896926171/work
RapidFuzz==3.14.3
readchar @ file:///C:/miniconda3/conda-bld/readchar_1760613474723/work
redis==7.1.0
referencing @ file:///C:/miniconda3/conda-bld/referencing_1762536906371/work
regex @ file:///C:/miniconda3/conda-bld/regex_1758887965793/work
requests==2.32.5
requests-file @ file:///C:/b/abs_804owi11qo/croot/requests-
file_1749719355197/work
```

```
requests-toolbelt @ file:///C:/Users/dev-admin/buildout/perseverance-python-
buildout/croot/requests-toolbelt_1731731341115/work
rfc3339_validator @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/rfc3339-validator_1729040163441/work
rfc3986_validator @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/rfc3986-validator_1729040192034/work
rich==13.9.4
roman-numerals-py @ file:///C:/b/abs_c6xpj9atdr/croot/roman-numerals-
py_1744666293821/work
rope @ file:///C:/miniconda3/conda-bld/rope_1762431497899/work
rpds-py @ file:///C:/miniconda3/conda-bld/rpds-py_1762366664211/work
rtree @ file:///C:/b/abs_66imkwbe1d/croot/rtree_1755874884341/work
ruamel.yaml @ file:///C:/miniconda3/conda-bld/ruamel.yaml_1762536064547/work
ruamel.yaml.clib @ file:///C:/miniconda3/conda-
bld/ruamel.yaml.clib_1762530094515/work
ruamel_yaml_conda @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/ruamel_yaml_1729063591675/work
ruff @ file:///C:/b/abs_6dm5slwq6g/croot/ruff_1750775626108/work
s3fs @ file:///C:/miniconda3/conda-bld/s3fs_1763105924370/work
scikit-image @ file:///C:/b/abs_10rg9ps1jh/croot/scikit-image_1750419495144/work
scikit-learn @ file:///C:/miniconda3/conda-bld/scikit-learn_1758619682247/work
scipy @ file:///C:/miniconda3/conda-bld/scipy_1762194988805/work/dist/scipy-
1.16.3-cp313-cp313-
win_amd64.whl#sha256=3cec51386c90cf8f9f27291f357385537f0c33b68c680a306702b0829fc
58fc2
Scrapy @ file:///C:/b/abs_e83a40s8op/croot/scrapy_1756709406800/work
seaborn @ file:///C:/b/abs_0bv5x0ucxb/croot/seaborn_1749110315600/work
semver @ file:///C:/miniconda3/conda-bld/semver_1761903323755/work
Send2Trash @ file:///C:/b/abs_7e73ol18dl/croot/send2trash_1736542724140/work
service-identity @
file:///C:/b/abs_2bq9_6ah3e/croot/service_identity_1749714991616/work
setuptools==80.9.0
shellingham @ file:///C:/miniconda3/conda-bld/shellingham_1761912227081/work
shiboken6==6.9.2
sip @ file:///C:/b/abs_f6ow8vlbb7/croot/sip_1756224172742/work
six==1.17.0
smmap @ file:///tmp/build/80754af9/smmap_1611694433573/work
sniffio==1.3.1
snowballstemmer @
file:///C:/b/abs_d7mwab58nt/croot/snowballstemmer_1756709980754/work
sortedcontainers @
file:///tmp/build/80754af9/sortedcontainers_1623949099177/work
soundcloud-v2==1.6.1
soupsieve==2.8.3
Sphinx @ file:///C:/b/abs_38umzzph7f/croot/sphinx_1744737150958/work
sphinxcontrib-applehelp @ file:///croot/sphinxcontrib-
applehelp_1744721330488/work
sphinxcontrib-devhelp @ file:///croot/sphinxcontrib-devhelp_1744727995579/work
```

```
sphinxcontrib-htmlhelp @ file:///croot/sphinxcontrib-htmlhelp_1744718370019/work
sphinxcontrib-jsmath @ file:///home/ktietz/src/ci/sphinxcontrib-
jsmath_1611920942228/work
sphinxcontrib-qthelp @ file:///croot/sphinxcontrib-qthelp_1744721051257/work
sphinxcontrib-serializinghtml @ file:///croot/sphinxcontrib-
serializinghtml_1744724097162/work
spotdl==4.4.3
spotipy==2.25.2
spyder @ file:///C:/miniconda3/conda-bld/spyder_1763761041704/work
spyder-kernels @ file:///C:/miniconda3/conda-bld/spyder-
kernels_1762167923331/work
SQLAlchemy @ file:///C:/b/abs_52irs3wj21/croot/sqlalchemy_1756710015498/work
stack_data @ file:///C:/miniconda3/conda-bld/stack_data_1757067051197/work
starlette==0.27.0
statsmodels @ file:///C:/b/abs_6f5u1m2ygp/croot/statsmodels_1753363817829/work
streamlit @ file:///C:/miniconda3/conda-bld/streamlit-
split_1761826278875/work/streamlit
superqt @ file:///C:/miniconda3/conda-bld/superqt_1758274748606/work
sympy @ file:///C:/b/abs_f1v7nwa6vp/croot/sympy_1756713483242/work
syncedlyrics==1.0.1
tables @ file:///C:/b/abs_e0h8gc4hz7/croot/pytables_1748452085563/work
tabulate @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/tabulate_1729056893645/work
tblib @ file:///C:/b/abs_a4ikhoig03/croot/tblib_1749739473644/work
tenacity @ file:///C:/miniconda3/conda-bld/tenacity_1760375744772/work
tensorflow==2.21.0
termcolor==3.3.0
terminado @ file:///C:/miniconda3/conda-bld/terminado_1757934993764/work
text-unidecode==1.3
textdistance @ file:///C:/b/abs_44khpg87dn/croot/textdistance_1751314517731/work
threadpoolctl @
file:///C:/b/abs_3904pspv77/croot/threadpoolctl_1731006105881/work
three-merge @ file:///tmp/build/80754af9/three-merge_1607553261110/work
tifffile @ file:///C:/miniconda3/conda-bld/tifffile_1759918264395/work
tinycss2 @ file:///C:/Users/dev-admin/buildout/croot/tinycss2_1739559171127/work
tldextract @ file:///C:/Users/dev-admin/buildout/perseverance-python-
buildout/croot/tldextract_1731748362836/work
toml @ file:///tmp/build/80754af9/toml_1616166611790/work
tomli @ file:///C:/b/abs_88e598m6o8/croot/tomli_1753774604115/work
tomlkit @ file:///C:/miniconda3/conda-bld/tomlkit_1762896656167/work
toolz @ file:///C:/b/abs_57avamxh36/croot/toolz_1736796279321/work
tornado @ file:///C:/b/abs_7dzpc171lf/croot/tornado_1748956950306/work
tqdm @ file:///C:/miniconda3/conda-bld/tqdm_1762863407729/work
traitlets @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/traitlets_1729038154689/work
truststore @ file:///C:/miniconda3/conda-bld/truststore_1762521027919/work
Twisted @ file:///C:/miniconda3/conda-bld/twisted_1762252507674/work
twisted-iocpsupport @ file:///C:/b/abs_6883jqhfrn/croot/twisted-
```

```
iocpsupport_1736544552637/work
typer==0.20.0
typer-slim==0.20.0
typing-inspection==0.4.2
typing_extensions==4.15.0
tzdata @ file:///croot/python-tzdata_1746123641790/work
uc-micro-py @ file:///C:/miniconda3/conda-bld/uc-micro-py_1758552131350/work
ujson @ file:///C:/miniconda3/conda-bld/ujson_1763119279088/work
Unidecode==1.4.0
urllib3==2.6.3
uvicorn==0.23.2
w3lib @ file:///C:/miniconda3/conda-bld/w3lib_1764240224379/work
watchdog @ file:///C:/miniconda3/conda-bld/watchdog_1759775798728/work
wcwidth @ file:///C:/b/abs_0c8p4c33bp/croot/wcwidth_1750352902378/work
webencodings @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/webencodings_1729039571130/work
websocket-client @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/websocket-client_1729041881346/work
websockets==14.2
Werkzeug @ file:///C:/b/abs_c7bupijx2_/croot/werkzeug_1737448709012/work
whatthepatch @ file:///C:/miniconda3/conda-bld/whatthepatch_1758276663508/work
wheel==0.45.1
widgetsnbextension @ file:///C:/miniconda3/conda-
bld/widgetsnbextension_1761556762350/work
win_inet_pton @ file:///C:/miniconda3/conda-bld/win_inet_pton_1761746278300/work
winloop @ file:///C:/miniconda3/conda-bld/winloop_1762178040759/work
wrapt==2.1.1
xarray @ file:///C:/miniconda3/conda-bld/xarray_1761080758145/work
xlwings @ file:///C:/miniconda3/conda-bld/xlwings_1758805172562/work
xyzservices @ file:///C:/miniconda3/conda-bld/xyzservices_1758720211246/work
yapf @ file:///C:/miniconda3/conda-bld/yapf_1761560513122/work
yarl @ file:///C:/miniconda3/conda-bld/yarl_1762250161944/work
yt-dlp==2026.1.31
ytmusicapi==1.11.5
zict @ file:///C:/Users/dev-admin/perseverance-python-
buildout/croot/zict_1729052457267/work
zipp @ file:///C:/miniconda3/conda-bld/zipp_1763977974806/work
zope.interface @ file:///C:/miniconda3/conda-
bld/zope.interface_1762337163127/work
zstandard @ file:///C:/miniconda3/conda-bld/zstandard_1758189089298/work
```

Hola voy a ejecutar el comando `!pip freeze` para cotillear a Google Colab

[3]:
```
!nvidia-smi
```

```
Sun Apr 12 23:29:38 2026
+-----------------------------------------------------------------------------
----------+
| NVIDIA-SMI 595.71                 Driver Version: 595.71         CUDA Version:
```

```
13.2     |
+-----------------------------------------+----------------------+------------
----------+
| GPU  Name                  Driver-Model | Bus-Id          Disp.A | Volatile
Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util
Compute M. |
|                                         |                      |
MIG M. |
|=========================================+======================+============
==========|
|   0  NVIDIA GeForce RTX 3060      WDDM  |   00000000:01:00.0  On |
N/A |
| 53%   46C    P8            N/A  /  N/A  |    1728MiB /  12288MiB |     18%
Default |
|                                         |                      |
N/A |
+-----------------------------------------+----------------------+------------
----------+

+-----------------------------------------------------------------------------
----------+
| Processes:
|
|  GPU   GI   CI              PID   Type   Process name
GPU Memory |
|        ID   ID
Usage      |
|=============================================================================
==========|
|    0   N/A  N/A            1436    C+G   C:\Program Files\LGHUB\lghub.exe
N/A      |
|    0   N/A  N/A            2720    C+G   C:\Windows\System32\dwm.exe
N/A      |
|    0   N/A  N/A            2764    C+G   …App_cw5n1h2txyewy\LockApp.exe
N/A      |
|    0   N/A  N/A            6296    C+G   …ends\LeagueClientUxRender.exe
N/A      |
|    0   N/A  N/A            8428    C+G   …SnippingTool\SnippingTool.exe
N/A      |
|    0   N/A  N/A            9524    C+G   …lpaper_engine\wallpaper64.exe
N/A      |
|    0   N/A  N/A           10832    C+G   …xyewy\ShellExperienceHost.exe
N/A      |
|    0   N/A  N/A           11760    C+G   …2txyewy\CrossDeviceResume.exe
N/A      |
|    0   N/A  N/A           11784    C+G   …Browser\Application\brave.exe
N/A      |
```

```
|    0   N/A  N/A            11800     C+G   …em_tray\lghub_system_tray.exe
N/A      |
|    0   N/A  N/A            11844     C+G   …n\NvContainer\nvcontainer.exe
N/A      |
|    0   N/A  N/A            13000     C+G   …IA App\CEF\NVIDIA Overlay.exe
N/A      |
|    0   N/A  N/A            15796     C+G   …IA App\CEF\NVIDIA Overlay.exe
N/A      |
|    0   N/A  N/A            17100     C+G   …8bbwe\PhoneExperienceHost.exe
N/A      |
|    0   N/A  N/A            17464     C+G   …0.3856.109\msedgewebview2.exe
N/A      |
|    0   N/A  N/A            17968     C+G   …crosoft shared\ink\TabTip.exe
N/A      |
|    0   N/A  N/A            18908     C+G   …5n1h2txyewy\TextInputHost.exe
N/A      |
|    0   N/A  N/A            19876     C+G   C:\Windows\explorer.exe
N/A      |
|    0   N/A  N/A            20464     C+G   …_cw5n1h2txyewy\SearchHost.exe
N/A      |
|    0   N/A  N/A            20476     C+G   …y\StartMenuExperienceHost.exe
N/A      |
|    0   N/A  N/A            20640     C+G   …indows\System32\ShellHost.exe
N/A      |
|    0   N/A  N/A            21348     C+G   …yb3d8bbwe\Notepad\Notepad.exe
N/A      |
|    0   N/A  N/A            22124     C+G   …cord\app-1.0.9231\Discord.exe
N/A      |
|    0   N/A  N/A            25228     C+G   …ms\Microsoft VS Code\Code.exe
N/A      |
|    0   N/A  N/A            27324     C+G   …Browser\Application\brave.exe
N/A      |
+--------------------------------------------------------------------------------
----------+
```

[4]:
```python
#Importemos TensorFlow 2.X y Numpy
import numpy as np
import tensorflow as tf
tf.__version__
```

[4]: '2.21.0'

- Cargando el conjunto de datos

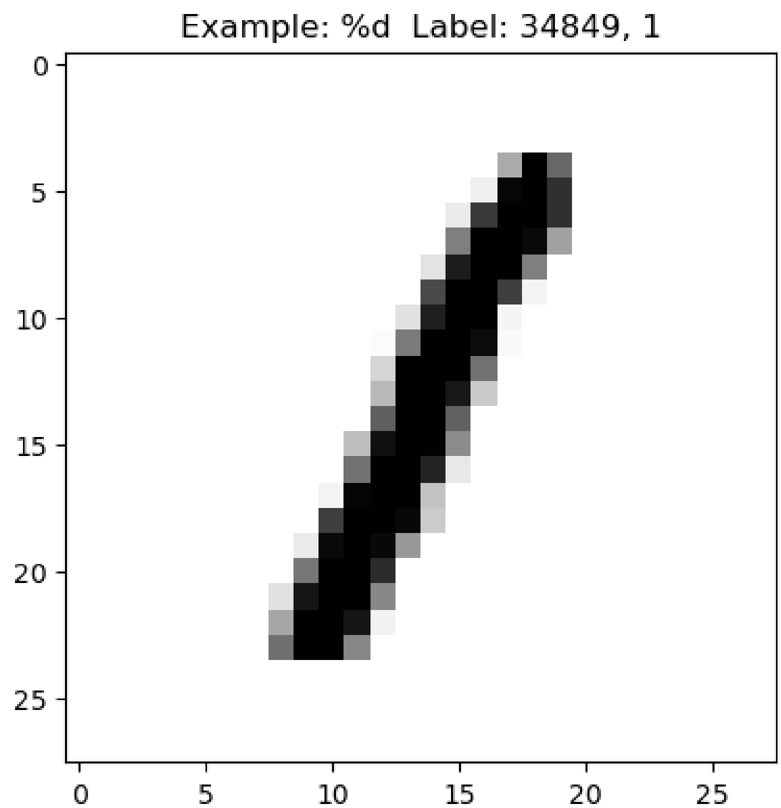
[5]:
```python
# Importamos el dataset MNIST y cargamos los datos
mnist = tf.keras.datasets.mnist
(x_train, y_train), (x_test, y_test) = mnist.load_data()
print(x_train.shape)
```

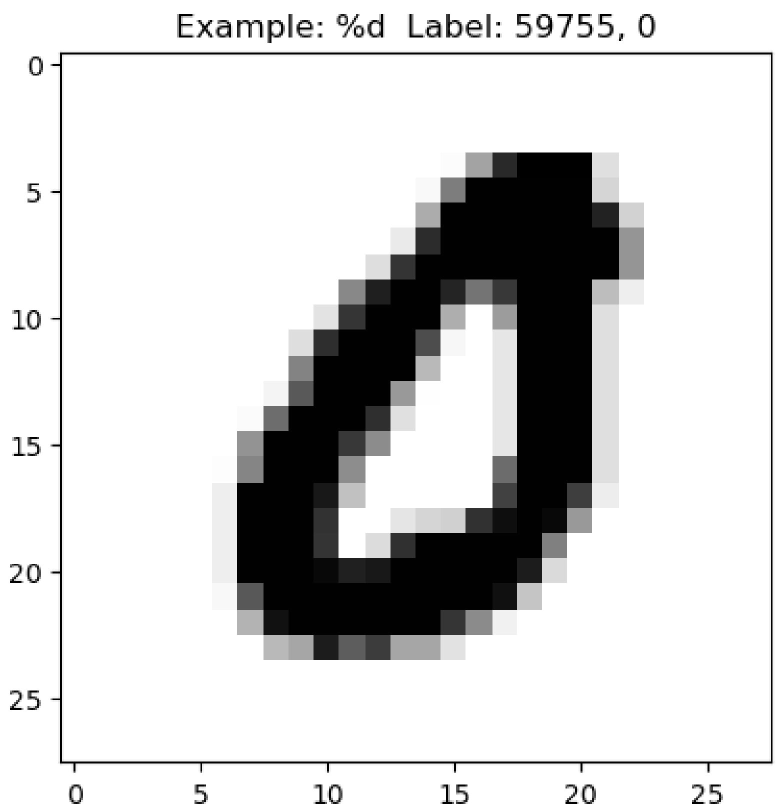
```
print(y_train.shape)
print(x_test.shape)
print(y_test.shape)
```

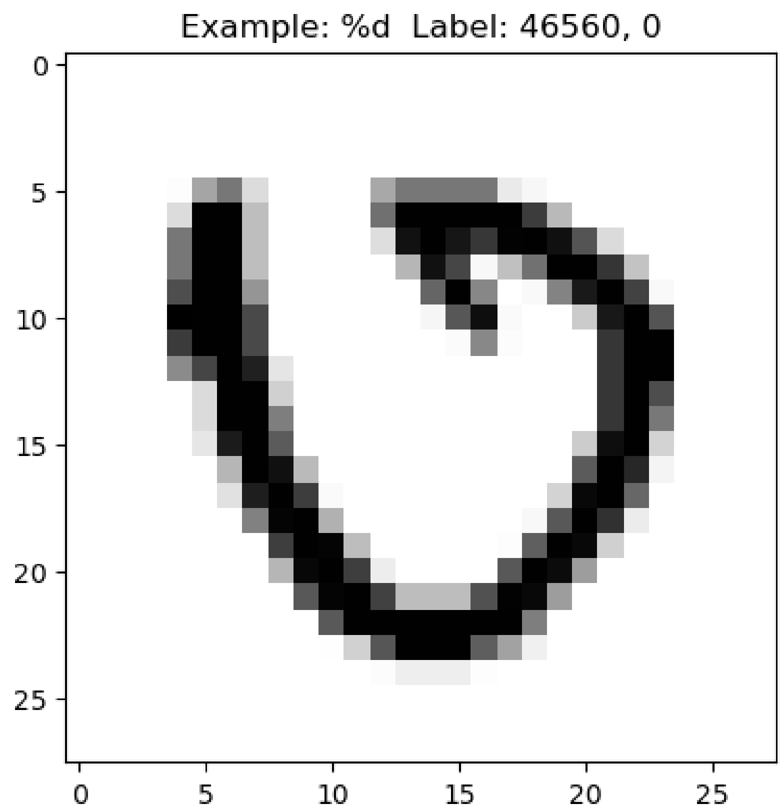
```
(60000, 28, 28)
(60000,)
(10000, 28, 28)
(10000,)
```

- Inspeccionando el conjunto de datos

[6]:
```
import matplotlib.pyplot as plt
# Función auxiliar para visualizar datos de entrenamiento de manera aleatoria
def display_digit(num,x_train,y_train):
  # Seleccionar la imagen num de mnist.train.images y hacer un reshape al␣
 ↪tamaño de la imagen
  image = x_train[num,:,:]
  # Seleccionar el target num de mnist.train.labels
  label = y_train[num]
  # Mostrar
  plt.title(f'Example: %d  Label: {num}, {label}')
  plt.imshow(image, cmap=plt.get_cmap('gray_r'))
  plt.show()

# Mostramos algunos ejemplos
display_digit(np.random.randint(0, x_train.shape[0]),x_train,y_train)
display_digit(np.random.randint(0, x_train.shape[0]),x_train,y_train)
display_digit(np.random.randint(0, x_train.shape[0]),x_train,y_train)
```


Example: %d Label: 34849, 1
0
5
10
15
20
25
0
5
10
15
20
25


Example: %d Label: 59755, 0
0
5
10
15
20
25
0
5
10
15
20
25



[7]:
```
np.random.randint(0, x_train.shape[0])
```

[7]: 23152

[ ]:
```

```

- Acondicionando el conjunto de datos

[8]:
```
# Pre-procesado obligatorio cuando trabajo con redes neuronales
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Dense, Flatten
from sklearn.model_selection import train_test_split

x_train, x_te = x_train / 255.0, x_test / 255.0 #Cambio al rango 0-1 ->
 ↪Disminuyo CC
y_train = to_categorical(y_train, num_classes=10) #One-hot encoding para
 ↪minimizar error
print(y_train.shape)
y_te = to_categorical(y_test, num_classes=10)
```

```
x_tr, x_val, y_tr, y_val = train_test_split(x_train, y_train, test_size=0.1,␣
  ↪random_state=42) # 3 subconjuntos es de vital importancia
print(y_tr.shape)
# print(y_tr[0])
```

```
(60000, 10)
(54000, 10)
```

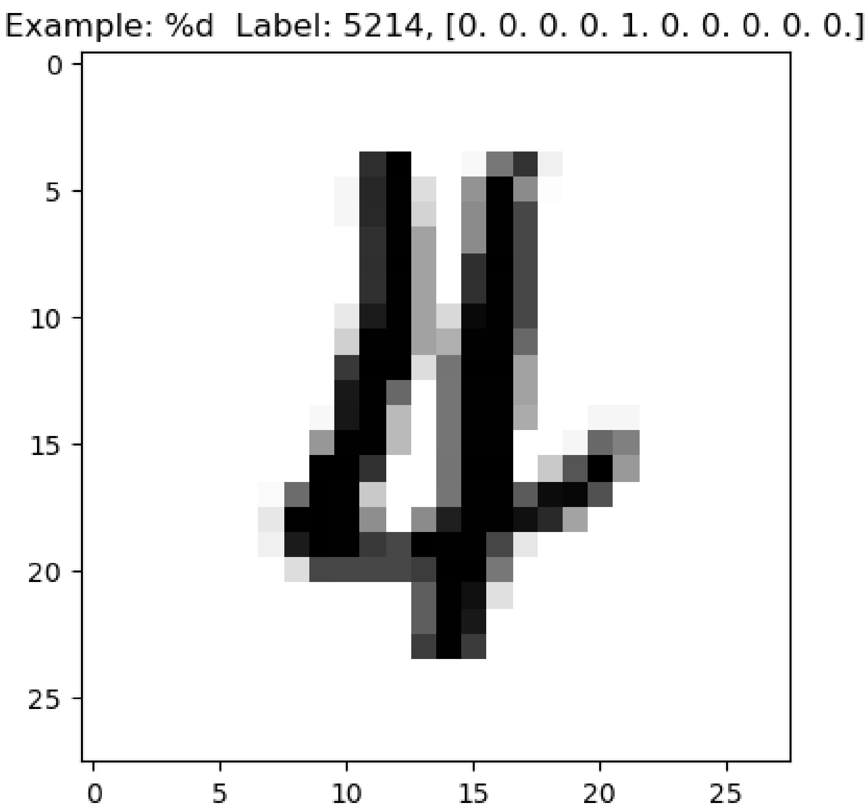
[9]:
```
print(y_tr[5214])
```

```
[0. 0. 0. 0. 1. 0. 0. 0. 0. 0.]
```

[10]:
```
print(y_tr[5214].argmax(axis=0))
```

```
4
```

[11]:
```
display_digit(5214, x_tr,y_tr)
```



- Creando la topología de Red Neuronal (MLP) y entrenándola

[12]:
```
# Voy a necesitar importar una serie de modulos para programar mi red neuronal
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, Input, Dropout
```

```python
# Vamos a codificar la topología de nuestra primera red neuronal!!!
model = Sequential()
model.add(Flatten(input_shape=(28,28)))
model.add(Dense(512, input_shape=(28*28,), activation="relu"))
model.add(Dense(10, activation="softmax")) #Capa salida -> Mismo número de␣
↪neuronas que de clases objetivo

# Ahí tenemos nuestro primer MLP con una única capa oculta de 512 neuronas
```

```
E:\anacconda\Lib\site-packages\keras\src\layers\reshaping\flatten.py:37:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
using Sequential models, prefer using an `Input(shape)` object as the first
layer in the model instead.
  super().__init__(**kwargs)
E:\anacconda\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using
Sequential models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

[13]:
```python
#Traigamos nuestro optimizador
from tensorflow.keras.optimizers import SGD
# Ahora que tengo definida la arquitectura, la compilo
model.compile(loss="categorical_crossentropy", optimizer=SGD(0.005),␣
↪metrics=["accuracy"])
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for
TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used.
Please use WSL2 or the TensorFlow-DirectML plugin.
```

[14]:
```python
######### Modelo funcional ############ (Clase 04/10/2023)
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Dense, Flatten, Dropout
from tensorflow.keras.optimizers import SGD

# Definir la entrada
input_layer = Input(shape=(28, 28))

# Capa de aplanamiento
x = Flatten()(input_layer)
x = Dense(512, activation="relu")(x)
# x = Dropout(0.5)(x)  # Introducción de capa de dropout en un modelo funcional
output_layer = Dense(10, activation="softmax")(x)

# Crear el modelo
model = Model(inputs=input_layer, outputs=output_layer)

# Compilar el modelo
```

```
model.compile(
    loss="categorical_crossentropy",
    optimizer=SGD(learning_rate=0.005),
    metrics=["accuracy"]
)
```

[15]:
```
print(model.summary())
```

Model: "functional_3"

```
 Layer (type)                         Output Shape                              ␣
 ↪Param #

 input_layer_1 (InputLayer)           (None, 28, 28)                                ␣
 ↪  0

 flatten_1 (Flatten)                  (None, 784)                                   ␣
 ↪  0

 dense_2 (Dense)                      (None, 512)                               ␣
 ↪401,920

 dense_3 (Dense)                      (None, 10)                                  ␣
 ↪5,130
```

Total params: 407,050 (1.55 MB)

Trainable params: 407,050 (1.55 MB)

Non-trainable params: 0 (0.00 B)

None

[16]:
```
# Por fin podemos entrenar nuestra primera red neuronal
print("[INFO]: Entrenando red neuronal...")
H = model.fit(x_tr, y_tr, validation_data=(x_val, y_val), epochs=50,␣
 ↪batch_size=128)
```

```
[INFO]: Entrenando red neuronal…
Epoch 1/50
422/422              5s 10ms/step -
accuracy: 0.6596 - loss: 1.5045 - val_accuracy: 0.8132 - val_loss: 0.9824
Epoch 2/50
```

```
422/422              4s 8ms/step -
accuracy: 0.8340 - loss: 0.7894 - val_accuracy: 0.8552 - val_loss: 0.6603
Epoch 3/50
422/422              4s 9ms/step -
accuracy: 0.8624 - loss: 0.5918 - val_accuracy: 0.8723 - val_loss: 0.5372
Epoch 4/50
422/422              4s 9ms/step -
accuracy: 0.8757 - loss: 0.5039 - val_accuracy: 0.8832 - val_loss: 0.4729
Epoch 5/50
422/422              3s 8ms/step -
accuracy: 0.8839 - loss: 0.4534 - val_accuracy: 0.8892 - val_loss: 0.4323
Epoch 6/50
422/422              4s 8ms/step -
accuracy: 0.8900 - loss: 0.4200 - val_accuracy: 0.8938 - val_loss: 0.4034
Epoch 7/50
422/422              3s 8ms/step -
accuracy: 0.8949 - loss: 0.3959 - val_accuracy: 0.8980 - val_loss: 0.3823
Epoch 8/50
422/422              3s 8ms/step -
accuracy: 0.8986 - loss: 0.3776 - val_accuracy: 0.9018 - val_loss: 0.3662
Epoch 9/50
422/422              5s 8ms/step -
accuracy: 0.9020 - loss: 0.3628 - val_accuracy: 0.9047 - val_loss: 0.3526
Epoch 10/50
422/422              4s 9ms/step -
accuracy: 0.9046 - loss: 0.3505 - val_accuracy: 0.9077 - val_loss: 0.3411
Epoch 11/50
422/422              6s 10ms/step -
accuracy: 0.9063 - loss: 0.3401 - val_accuracy: 0.9105 - val_loss: 0.3312
Epoch 12/50
422/422              4s 9ms/step -
accuracy: 0.9088 - loss: 0.3310 - val_accuracy: 0.9115 - val_loss: 0.3230
Epoch 13/50
422/422              5s 8ms/step -
accuracy: 0.9105 - loss: 0.3229 - val_accuracy: 0.9133 - val_loss: 0.3154
Epoch 14/50
422/422              3s 8ms/step -
accuracy: 0.9123 - loss: 0.3157 - val_accuracy: 0.9150 - val_loss: 0.3084
Epoch 15/50
422/422              3s 7ms/step -
accuracy: 0.9142 - loss: 0.3091 - val_accuracy: 0.9173 - val_loss: 0.3029
Epoch 16/50
422/422              4s 9ms/step -
accuracy: 0.9157 - loss: 0.3030 - val_accuracy: 0.9177 - val_loss: 0.2965
Epoch 17/50
422/422              4s 9ms/step -
accuracy: 0.9171 - loss: 0.2974 - val_accuracy: 0.9185 - val_loss: 0.2916
Epoch 18/50
```

```
422/422              4s 10ms/step -
accuracy: 0.9192 - loss: 0.2922 - val_accuracy: 0.9213 - val_loss: 0.2866
Epoch 19/50
422/422              5s 10ms/step -
accuracy: 0.9208 - loss: 0.2873 - val_accuracy: 0.9218 - val_loss: 0.2819
Epoch 20/50
422/422              4s 10ms/step -
accuracy: 0.9216 - loss: 0.2828 - val_accuracy: 0.9240 - val_loss: 0.2776
Epoch 21/50
422/422              4s 10ms/step -
accuracy: 0.9229 - loss: 0.2784 - val_accuracy: 0.9248 - val_loss: 0.2735
Epoch 22/50
422/422              4s 10ms/step -
accuracy: 0.9243 - loss: 0.2743 - val_accuracy: 0.9257 - val_loss: 0.2698
Epoch 23/50
422/422              5s 9ms/step -
accuracy: 0.9255 - loss: 0.2704 - val_accuracy: 0.9273 - val_loss: 0.2660
Epoch 24/50
422/422              4s 10ms/step -
accuracy: 0.9266 - loss: 0.2666 - val_accuracy: 0.9285 - val_loss: 0.2626
Epoch 25/50
422/422              5s 9ms/step -
accuracy: 0.9274 - loss: 0.2631 - val_accuracy: 0.9288 - val_loss: 0.2592
Epoch 26/50
422/422              5s 11ms/step -
accuracy: 0.9286 - loss: 0.2595 - val_accuracy: 0.9287 - val_loss: 0.2563
Epoch 27/50
422/422              5s 10ms/step -
accuracy: 0.9298 - loss: 0.2562 - val_accuracy: 0.9295 - val_loss: 0.2529
Epoch 28/50
422/422              5s 8ms/step -
accuracy: 0.9305 - loss: 0.2531 - val_accuracy: 0.9298 - val_loss: 0.2501
Epoch 29/50
422/422              3s 8ms/step -
accuracy: 0.9314 - loss: 0.2499 - val_accuracy: 0.9313 - val_loss: 0.2475
Epoch 30/50
422/422              6s 9ms/step -
accuracy: 0.9321 - loss: 0.2469 - val_accuracy: 0.9325 - val_loss: 0.2445
Epoch 31/50
422/422              4s 8ms/step -
accuracy: 0.9330 - loss: 0.2440 - val_accuracy: 0.9328 - val_loss: 0.2415
Epoch 32/50
422/422              3s 8ms/step -
accuracy: 0.9337 - loss: 0.2412 - val_accuracy: 0.9332 - val_loss: 0.2387
Epoch 33/50
422/422              5s 8ms/step -
accuracy: 0.9346 - loss: 0.2384 - val_accuracy: 0.9347 - val_loss: 0.2363
Epoch 34/50
```

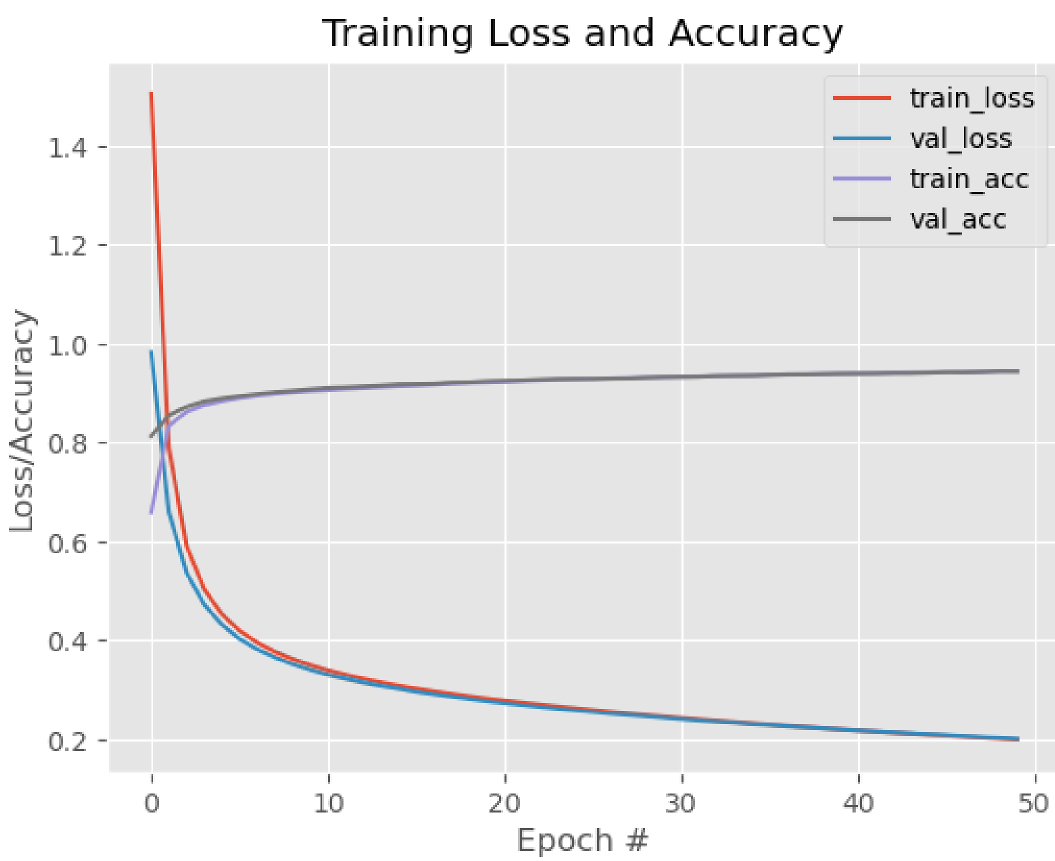
```
422/422              4s 8ms/step -
accuracy: 0.9348 - loss: 0.2356 - val_accuracy: 0.9353 - val_loss: 0.2341
Epoch 35/50
422/422              3s 8ms/step -
accuracy: 0.9358 - loss: 0.2330 - val_accuracy: 0.9353 - val_loss: 0.2312
Epoch 36/50
422/422              3s 7ms/step -
accuracy: 0.9369 - loss: 0.2305 - val_accuracy: 0.9367 - val_loss: 0.2295
Epoch 37/50
422/422              3s 8ms/step -
accuracy: 0.9374 - loss: 0.2281 - val_accuracy: 0.9378 - val_loss: 0.2268
Epoch 38/50
422/422              3s 8ms/step -
accuracy: 0.9378 - loss: 0.2257 - val_accuracy: 0.9378 - val_loss: 0.2247
Epoch 39/50
422/422              4s 8ms/step -
accuracy: 0.9385 - loss: 0.2233 - val_accuracy: 0.9385 - val_loss: 0.2228
Epoch 40/50
422/422              3s 8ms/step -
accuracy: 0.9392 - loss: 0.2210 - val_accuracy: 0.9392 - val_loss: 0.2204
Epoch 41/50
422/422              4s 9ms/step -
accuracy: 0.9397 - loss: 0.2187 - val_accuracy: 0.9393 - val_loss: 0.2184
Epoch 42/50
422/422              5s 8ms/step -
accuracy: 0.9403 - loss: 0.2165 - val_accuracy: 0.9395 - val_loss: 0.2163
Epoch 43/50
422/422              3s 8ms/step -
accuracy: 0.9411 - loss: 0.2143 - val_accuracy: 0.9402 - val_loss: 0.2146
Epoch 44/50
422/422              4s 8ms/step -
accuracy: 0.9415 - loss: 0.2122 - val_accuracy: 0.9402 - val_loss: 0.2126
Epoch 45/50
422/422              4s 8ms/step -
accuracy: 0.9420 - loss: 0.2101 - val_accuracy: 0.9410 - val_loss: 0.2108
Epoch 46/50
422/422              4s 8ms/step -
accuracy: 0.9424 - loss: 0.2081 - val_accuracy: 0.9423 - val_loss: 0.2091
Epoch 47/50
422/422              5s 8ms/step -
accuracy: 0.9433 - loss: 0.2061 - val_accuracy: 0.9418 - val_loss: 0.2071
Epoch 48/50
422/422              3s 8ms/step -
accuracy: 0.9439 - loss: 0.2041 - val_accuracy: 0.9425 - val_loss: 0.2054
Epoch 49/50
422/422              4s 9ms/step -
accuracy: 0.9444 - loss: 0.2021 - val_accuracy: 0.9440 - val_loss: 0.2040
Epoch 50/50
```

```
422/422              4s 8ms/step -
accuracy: 0.9448 - loss: 0.2003 - val_accuracy: 0.9437 - val_loss: 0.2020
```

- Observando el proceso de entrenamiento para tomar decisiones

[17]:
```python
# Muestro gráfica de accuracy y losses
plt.style.use("ggplot")
plt.figure()
plt.plot(np.arange(0, 50), H.history["loss"], label="train_loss")
plt.plot(np.arange(0, 50), H.history["val_loss"], label="val_loss")
plt.plot(np.arange(0, 50), H.history["accuracy"], label="train_acc")
plt.plot(np.arange(0, 50), H.history["val_accuracy"], label="val_acc")
plt.title("Training Loss and Accuracy")
plt.xlabel("Epoch #")
plt.ylabel("Loss/Accuracy")
plt.legend()
```

[17]:
```
<matplotlib.legend.Legend at 0x21318c50ad0>
```



- Probando el conjunto de datos en el subset de test y evaluando el performance del modelo

[18]:
```
from sklearn.metrics import classification_report
# Evaluando el modelo de predicción con las imágenes de test
print("[INFO]: Evaluando red neuronal...")
predictions = model.predict(x_te, batch_size=128)
print(y_te[0])
print(predictions[0])
print(classification_report(y_te.argmax(axis=1), predictions.argmax(axis=1)))
```

```
[INFO]: Evaluando red neuronal…
79/79             1s 6ms/step
[0. 0. 0. 0. 0. 0. 0. 1. 0. 0.]
[9.57644443e-05 2.44640916e-07 5.41098765e-04 2.44255015e-03
 2.05375773e-06 1.09680404e-04 2.84212422e-08 9.96462524e-01
 1.96422534e-05 3.26442503e-04]
              precision    recall  f1-score   support

           0       0.96      0.98      0.97       980
           1       0.98      0.98      0.98      1135
           2       0.95      0.93      0.94      1032
           3       0.92      0.94      0.93      1010
           4       0.94      0.95      0.94       982
           5       0.94      0.92      0.93       892
           6       0.95      0.96      0.95       958
           7       0.94      0.93      0.94      1028
           8       0.93      0.93      0.93       974
           9       0.94      0.92      0.93      1009

    accuracy                           0.94     10000
   macro avg       0.94      0.94      0.94     10000
weighted avg       0.94      0.94      0.94     10000
```

[19]:
```
predictions[0].argmax(axis=0)
```

[19]: np.int64(7)

[20]:
```
y_te[0].argmax(axis=0)
```

[20]: np.int64(7)

[21]:
```
print(model.summary())
```

```
Model: "functional_3"


 Layer (type)                         Output Shape                         ␣
↪Param #
```

```
 input_layer_1 (InputLayer)            (None, 28, 28)                          ␣
↪  0

 flatten_1 (Flatten)                   (None, 784)                             ␣
↪  0

 dense_2 (Dense)                       (None, 512)                         ␣
↪401,920

 dense_3 (Dense)                       (None, 10)                            ␣
↪5,130
```

Total params: 407,052 (1.55 MB)

Trainable params: 407,050 (1.55 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 2 (12.00 B)

```
None
```

[22]:
```
model_mnist = model
```

0.2 MLP APLICADO A TEXTO: EJEMPLO REUTERS

- Cargando el conjunto de datos

[23]:
```
import numpy as np
import tensorflow as tf
# Importamos el dataset REUTERS y cargamos los datos
reuters = tf.keras.datasets.reuters
WORD_LIMIT = 10000
(training_data, training_labels), (testing_data, testing_labels) = reuters.
 ↪load_data(num_words=WORD_LIMIT)
print(training_data.shape)
print(training_labels.shape)
print(testing_data.shape)
print(testing_labels.shape)
```

```
(8982,)
(8982,)
(2246,)
(2246,)
```

- Inspeccionando el conjunto de datos

[24]:
```
# Los datos son numericos para decodificarlos, se puede usar reuters.
 ↪get_word_index()
word_index = reuters.get_word_index()
reverse_word_index = dict({value : key for key, value in word_index.items()})
decoded = ' '.join(
    [reverse_word_index.get(i-3,'?') for i in training_data[5]]
)
decoded
```

[24]:
```
"? the u s agriculture department estimated canada's 1986 87 wheat crop at 31 85
mln tonnes vs 31 85 mln tonnes last month it estimated 1985 86 output at 24 25
mln tonnes vs 24 25 mln last month canadian 1986 87 coarse grain production is
projected at 27 62 mln tonnes vs 27 62 mln tonnes last month production in 1985
86 is estimated at 24 95 mln tonnes vs 24 95 mln last month canadian wheat
exports in 1986 87 are forecast at 19 00 mln tonnes vs 18 00 mln tonnes last
month exports in 1985 86 are estimated at 17 71 mln tonnes vs 17 72 mln last
month reuter 3"
```

- Acondicionando el conjunto de datos

[25]:
```
# Función auxiliar para representar las palabras (que no entiende nuestra red␣
 ↪neuronal) en números
import numpy as np
# one hot encoding del input, vector con cada indice indicando si una palabra␣
 ↪esta presente
def one_hot_encode(data):
    encoded = np.zeros((len(data),WORD_LIMIT))
    for i, v in enumerate(data):
        encoded[i,v] = 1 # localiza las columnas del genero correspondiente,␣
 ↪marca con 1
    return encoded
```

[26]:
```
# Convertimos nuestras palabras a números
x_train = one_hot_encode(training_data)
x_test = one_hot_encode(testing_data)
print(x_train.shape)
print(x_test.shape)
```

```
(8982, 10000)
(2246, 10000)
```

[27]:
```
print(x_test[3])
```

```
[0. 1. 1. … 0. 0. 0.]
```

[28]:
```
# Convertimos nuestros labels (categoría reseña) a one-hot encoding
from tensorflow.keras.utils import to_categorical
y_train = to_categorical(training_labels)
```

```
y_test = to_categorical(testing_labels)
print(y_train.shape)
print(y_test.shape)
```

```
(8982, 46)
(2246, 46)
```

[29]:
```
print(y_test[3])
```

```
[0. 0. 0. 0. 1. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
```

- Creando la topología de Red Neuronal (MLP) y entrenándola

[30]:
```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
# Vamos a codificar la topología de nuestro MLP
model = Sequential()
model.add(Dense(128,activation='relu',input_shape=(WORD_LIMIT,)))
model.add(Dense(64,activation='relu'))
model.add(Dense(46,activation='softmax')) # Reparto de la unidad de␣
 ↪probabilidad entre num_classes
# En el caso de regresión cambiamos la ultima capa por:
#model.add(Dense(1,activation='sigmoid'))
```

```
E:\anacconda\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using
Sequential models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

[31]:
```
# Ahora que tengo definida la arquitectura, la compilo
model.compile(optimizer='adam',
              loss='categorical_crossentropy', # ideal para clasificacion␣
 ↪multiclase/ para regresion (mse/mae)
              metrics=['accuracy'])
```

[32]:
```
# A entrenar nuestra red neuronal sea dicho!
H = model.fit(x_train,y_train,epochs=20,batch_size=32, validation_split=0.2)
```

```
Epoch 1/20
225/225             6s 18ms/step -
accuracy: 0.6614 - loss: 1.5048 - val_accuracy: 0.7607 - val_loss: 1.0320
Epoch 2/20
225/225             5s 16ms/step -
accuracy: 0.8605 - loss: 0.6022 - val_accuracy: 0.7991 - val_loss: 0.8709
Epoch 3/20
225/225             4s 17ms/step -
accuracy: 0.9365 - loss: 0.2851 - val_accuracy: 0.8063 - val_loss: 0.8942
Epoch 4/20
```

```
225/225              6s 19ms/step -
accuracy: 0.9538 - loss: 0.1893 - val_accuracy: 0.8136 - val_loss: 0.9055
Epoch 5/20
225/225              5s 16ms/step -
accuracy: 0.9571 - loss: 0.1571 - val_accuracy: 0.7986 - val_loss: 0.9669
Epoch 6/20
225/225              4s 17ms/step -
accuracy: 0.9595 - loss: 0.1382 - val_accuracy: 0.7896 - val_loss: 1.0618
Epoch 7/20
225/225              4s 19ms/step -
accuracy: 0.9608 - loss: 0.1240 - val_accuracy: 0.7997 - val_loss: 1.0386
Epoch 8/20
225/225              5s 17ms/step -
accuracy: 0.9596 - loss: 0.1168 - val_accuracy: 0.8036 - val_loss: 1.0430
Epoch 9/20
225/225              4s 17ms/step -
accuracy: 0.9610 - loss: 0.1112 - val_accuracy: 0.7963 - val_loss: 1.0371
Epoch 10/20
225/225              4s 17ms/step -
accuracy: 0.9608 - loss: 0.1059 - val_accuracy: 0.7958 - val_loss: 1.1275
Epoch 11/20
225/225              4s 18ms/step -
accuracy: 0.9614 - loss: 0.1005 - val_accuracy: 0.8008 - val_loss: 1.0963
Epoch 12/20
225/225              5s 18ms/step -
accuracy: 0.9613 - loss: 0.0977 - val_accuracy: 0.7986 - val_loss: 1.1536
Epoch 13/20
225/225              4s 17ms/step -
accuracy: 0.9603 - loss: 0.0927 - val_accuracy: 0.7908 - val_loss: 1.2038
Epoch 14/20
225/225              6s 19ms/step -
accuracy: 0.9602 - loss: 0.0921 - val_accuracy: 0.7885 - val_loss: 1.1733
Epoch 15/20
225/225              5s 17ms/step -
accuracy: 0.9624 - loss: 0.0837 - val_accuracy: 0.7835 - val_loss: 1.2322
Epoch 16/20
225/225              4s 18ms/step -
accuracy: 0.9619 - loss: 0.0825 - val_accuracy: 0.7986 - val_loss: 1.1479
Epoch 17/20
225/225              4s 17ms/step -
accuracy: 0.9612 - loss: 0.0805 - val_accuracy: 0.7896 - val_loss: 1.2322
Epoch 18/20
225/225              5s 16ms/step -
accuracy: 0.9621 - loss: 0.0799 - val_accuracy: 0.7869 - val_loss: 1.2131
Epoch 19/20
225/225              4s 17ms/step -
accuracy: 0.9610 - loss: 0.0774 - val_accuracy: 0.7919 - val_loss: 1.2535
Epoch 20/20
```

```
225/225              5s 19ms/step -
accuracy: 0.9617 - loss: 0.0755 - val_accuracy: 0.7819 - val_loss: 1.3495
```

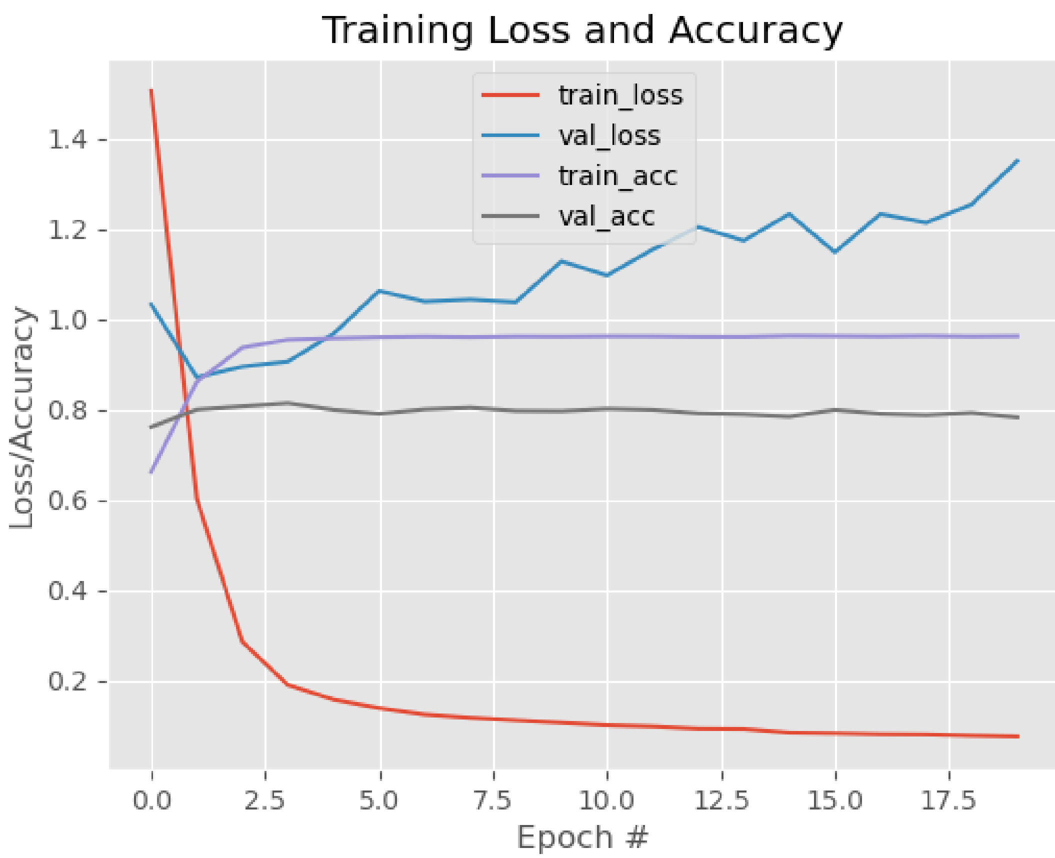
- Observando el proceso de entrenamiento para tomar decisiones

[33]:
```python
model.summary()
```

```
Model: "sequential_1"


 Layer (type)                          Output Shape                              ␣
↪Param #

 dense_4 (Dense)                       (None, 128)                             ␣
↪1,280,128

 dense_5 (Dense)                       (None, 64)                                ␣
↪8,256

 dense_6 (Dense)                       (None, 46)                                ␣
↪2,990



 Total params: 3,874,124 (14.78 MB)


 Trainable params: 1,291,374 (4.93 MB)


 Non-trainable params: 0 (0.00 B)


 Optimizer params: 2,582,750 (9.85 MB)
```

[34]:
```python
import matplotlib.pyplot as plt
# Muestro gráfica de accuracy y losses
plt.style.use("ggplot")
plt.figure()
plt.plot(np.arange(0, 20), H.history["loss"], label="train_loss")
plt.plot(np.arange(0, 20), H.history["val_loss"], label="val_loss")
plt.plot(np.arange(0, 20), H.history["accuracy"], label="train_acc")
plt.plot(np.arange(0, 20), H.history["val_accuracy"], label="val_acc")
plt.title("Training Loss and Accuracy")
plt.xlabel("Epoch #")
plt.ylabel("Loss/Accuracy")
plt.legend()
```

[34]:
```
<matplotlib.legend.Legend at 0x2134fc4f390>
```



- Probando el conjunto de datos en el subset de test y evaluando el performance del modelo

[35]:
```
# Evaluando el modelo de predicción con las imágenes de test
print("[INFO]: Evaluando red neuronal...")
model.predict(x_test)
loss, accuracy = model.evaluate(x_test, y_test)
print('Loss {}, accuracy {}'.format(loss,accuracy))
```

```
[INFO]: Evaluando red neuronal…
71/71              0s 5ms/step
71/71              0s 5ms/step -
accuracy: 0.7809 - loss: 1.3969
Loss 1.3969472646713257, accuracy 0.7809438705444336
```

0.3 REGULARIZACIÓN EN APRENDIZAJE PROFUNDO

- Weight regularization L1/L2

[36]:
```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras import regularizers
```

```
# Vamos a codificar la topología de nuestro MLP
model_reg = Sequential()
model_reg.add(Dense(128,activation='relu', kernel_regularizer=regularizers.l2(0.
 ↪01), input_shape=(WORD_LIMIT,)))
model_reg.add(Dense(64,activation='relu', kernel_regularizer=regularizers.l2(0.
 ↪01)))
model_reg.add(Dense(46,activation='softmax')) # Reparto de la unidad de␣
 ↪probabilidad entre num_classes
```

```
E:\anacconda\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using
Sequential models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

[37]:
```
# Ahora que tengo definida la arquitectura, la compilo
model_reg.compile(optimizer='adam',
              loss='categorical_crossentropy', # ideal para clasificacion␣
 ↪multiclase
              metrics=['accuracy'])
```

[38]:
```
# A entrenar nuestra red neuronal sea dicho!
H = model_reg.fit(x_train,y_train,epochs=20,batch_size=32, validation_split=0.2)
```

```
Epoch 1/20
225/225          7s 20ms/step -
accuracy: 0.6505 - loss: 2.4017 - val_accuracy: 0.7223 - val_loss: 1.8214
Epoch 2/20
225/225          5s 21ms/step -
accuracy: 0.7350 - loss: 1.6997 - val_accuracy: 0.7435 - val_loss: 1.6900
Epoch 3/20
225/225          5s 22ms/step -
accuracy: 0.7581 - loss: 1.5899 - val_accuracy: 0.7551 - val_loss: 1.6258
Epoch 4/20
225/225          5s 22ms/step -
accuracy: 0.7688 - loss: 1.5293 - val_accuracy: 0.7401 - val_loss: 1.5982
Epoch 5/20
225/225          4s 19ms/step -
accuracy: 0.7761 - loss: 1.5005 - val_accuracy: 0.7602 - val_loss: 1.5696
Epoch 6/20
225/225          5s 21ms/step -
accuracy: 0.7820 - loss: 1.4605 - val_accuracy: 0.7546 - val_loss: 1.5821
Epoch 7/20
225/225          5s 20ms/step -
accuracy: 0.7848 - loss: 1.4280 - val_accuracy: 0.7718 - val_loss: 1.5205
Epoch 8/20
225/225          4s 19ms/step -
accuracy: 0.7958 - loss: 1.3990 - val_accuracy: 0.7524 - val_loss: 1.5507
```

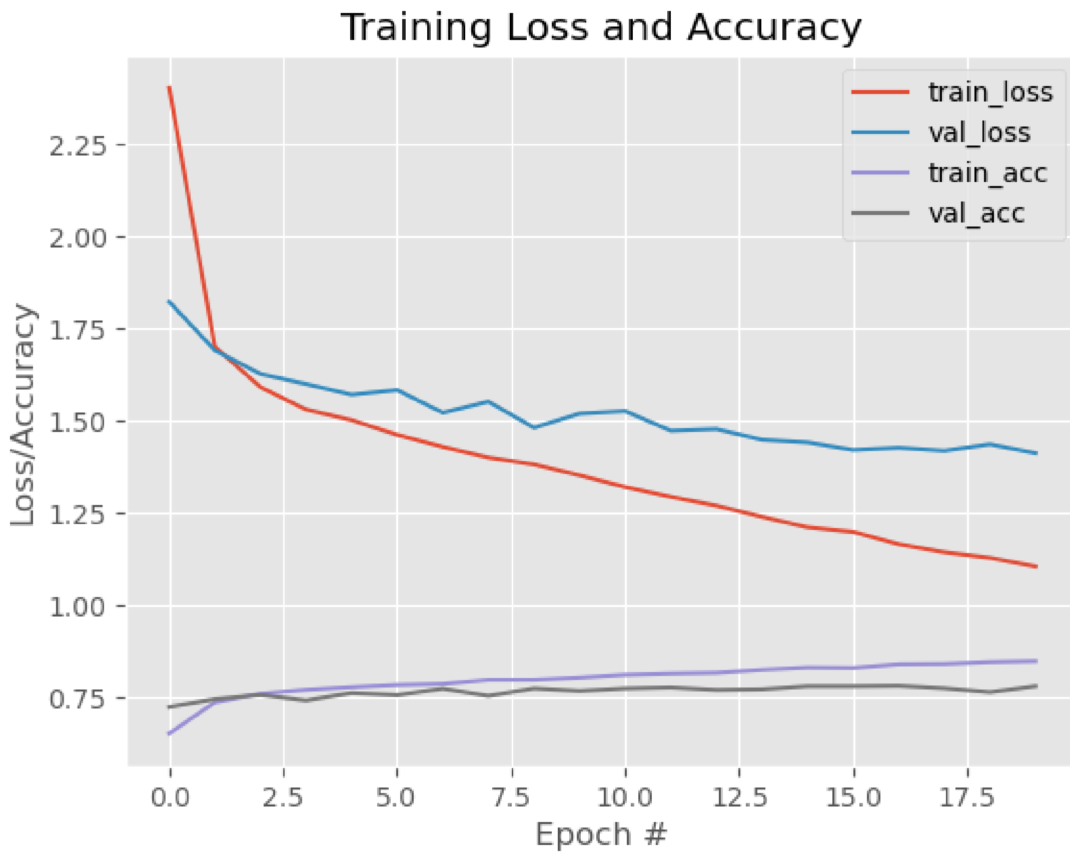
```
Epoch 9/20
225/225              5s 21ms/step -
accuracy: 0.7958 - loss: 1.3803 - val_accuracy: 0.7724 - val_loss: 1.4799
Epoch 10/20
225/225              5s 20ms/step -
accuracy: 0.8018 - loss: 1.3507 - val_accuracy: 0.7657 - val_loss: 1.5182
Epoch 11/20
225/225              4s 19ms/step -
accuracy: 0.8095 - loss: 1.3190 - val_accuracy: 0.7724 - val_loss: 1.5249
Epoch 12/20
225/225              6s 21ms/step -
accuracy: 0.8128 - loss: 1.2926 - val_accuracy: 0.7752 - val_loss: 1.4718
Epoch 13/20
225/225              5s 20ms/step -
accuracy: 0.8149 - loss: 1.2686 - val_accuracy: 0.7685 - val_loss: 1.4762
Epoch 14/20
225/225              5s 20ms/step -
accuracy: 0.8231 - loss: 1.2380 - val_accuracy: 0.7702 - val_loss: 1.4472
Epoch 15/20
225/225              5s 21ms/step -
accuracy: 0.8291 - loss: 1.2099 - val_accuracy: 0.7785 - val_loss: 1.4401
Epoch 16/20
225/225              5s 19ms/step -
accuracy: 0.8281 - loss: 1.1972 - val_accuracy: 0.7785 - val_loss: 1.4196
Epoch 17/20
225/225              5s 21ms/step -
accuracy: 0.8381 - loss: 1.1639 - val_accuracy: 0.7796 - val_loss: 1.4256
Epoch 18/20
225/225              5s 20ms/step -
accuracy: 0.8390 - loss: 1.1421 - val_accuracy: 0.7730 - val_loss: 1.4175
Epoch 19/20
225/225              5s 20ms/step -
accuracy: 0.8440 - loss: 1.1270 - val_accuracy: 0.7624 - val_loss: 1.4349
Epoch 20/20
225/225              5s 21ms/step -
accuracy: 0.8463 - loss: 1.1038 - val_accuracy: 0.7785 - val_loss: 1.4109
```

[39]:

```python
import matplotlib.pyplot as plt
# Muestro gráfica de accuracy y losses
plt.style.use("ggplot")
plt.figure()
plt.plot(np.arange(0, 20), H.history["loss"], label="train_loss")
plt.plot(np.arange(0, 20), H.history["val_loss"], label="val_loss")
plt.plot(np.arange(0, 20), H.history["accuracy"], label="train_acc")
plt.plot(np.arange(0, 20), H.history["val_accuracy"], label="val_acc")
plt.title("Training Loss and Accuracy")
plt.xlabel("Epoch #")
```

```
plt.ylabel("Loss/Accuracy")
plt.legend()
```

[39]: <matplotlib.legend.Legend at 0x2134c49a990>



[40]:
```
# Evaluando el modelo de predicción con las imágenes de test
print("[INFO]: Evaluando red neuronal...")
model_reg.predict(x_test)
loss, accuracy = model_reg.evaluate(x_test, y_test)
print('Loss {}, accuracy {}'.format(loss,accuracy))
```

```
[INFO]: Evaluando red neuronal…
71/71             0s 5ms/step
71/71             1s 7ms/step -
accuracy: 0.7645 - loss: 1.4571
Loss 1.457085132598877, accuracy 0.7644701600074768
```

- Dropout

[41]:
```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
# Vamos a codificar la topología de nuestro MLP
model_drop = Sequential()
model_drop.add(Dense(128,activation='relu', input_shape=(WORD_LIMIT,)))
#model_drop.add(Dropout(0.5))
model_drop.add(Dropout(0.55))
model_drop.add(Dense(64,activation='relu'))
#model_drop.add(Dropout(0.5))
model_drop.add(Dropout(0.55))
model_drop.add(Dense(46,activation='softmax')) # Reparto de la unidad de␣
↪probabilidad entre num_classes
```

```
E:\anacconda\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using
Sequential models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

[42]:
```
# Ahora que tengo definida la arquitectura, la compilo
model_drop.compile(optimizer='adam',
               loss='categorical_crossentropy', # ideal para clasificacion␣
↪multiclase
               metrics=['accuracy'])
```

[43]:
```
# A entrenar nuestra red neuronal sea dicho!
H = model_drop.fit(x_train,y_train,epochs=20,batch_size=32, validation_split=0.
↪2)
```

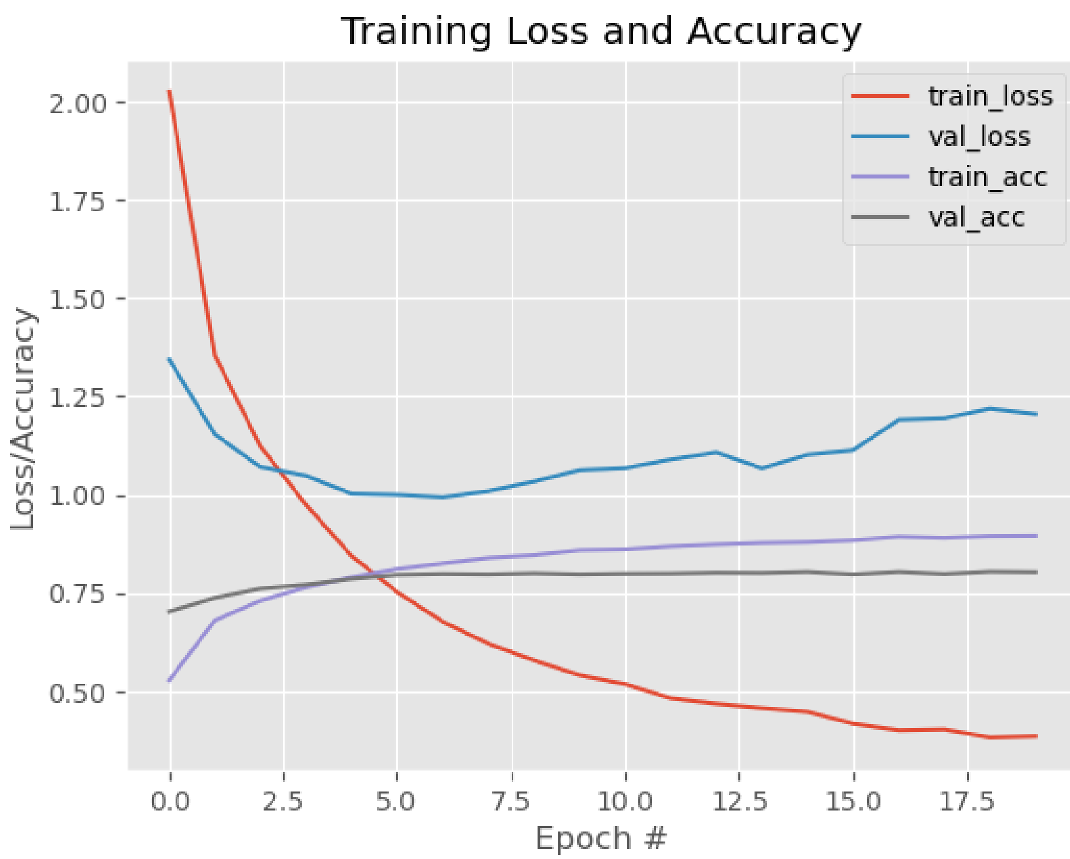
```
Epoch 1/20
225/225            6s 18ms/step -
accuracy: 0.5276 - loss: 2.0250 - val_accuracy: 0.7023 - val_loss: 1.3435
Epoch 2/20
225/225            4s 15ms/step -
accuracy: 0.6797 - loss: 1.3543 - val_accuracy: 0.7373 - val_loss: 1.1536
Epoch 3/20
225/225            4s 18ms/step -
accuracy: 0.7301 - loss: 1.1231 - val_accuracy: 0.7607 - val_loss: 1.0703
Epoch 4/20
225/225            4s 16ms/step -
accuracy: 0.7641 - loss: 0.9743 - val_accuracy: 0.7707 - val_loss: 1.0481
Epoch 5/20
225/225            4s 17ms/step -
accuracy: 0.7897 - loss: 0.8438 - val_accuracy: 0.7858 - val_loss: 1.0024
Epoch 6/20
225/225            4s 17ms/step -
accuracy: 0.8109 - loss: 0.7520 - val_accuracy: 0.7958 - val_loss: 0.9998
Epoch 7/20
```

```
225/225              4s 18ms/step -
accuracy: 0.8249 - loss: 0.6765 - val_accuracy: 0.7980 - val_loss: 0.9933
Epoch 8/20
225/225              6s 19ms/step -
accuracy: 0.8388 - loss: 0.6207 - val_accuracy: 0.7969 - val_loss: 1.0090
Epoch 9/20
225/225              4s 17ms/step -
accuracy: 0.8459 - loss: 0.5782 - val_accuracy: 0.7997 - val_loss: 1.0335
Epoch 10/20
225/225              4s 18ms/step -
accuracy: 0.8587 - loss: 0.5406 - val_accuracy: 0.7969 - val_loss: 1.0621
Epoch 11/20
225/225              4s 18ms/step -
accuracy: 0.8607 - loss: 0.5176 - val_accuracy: 0.7986 - val_loss: 1.0672
Epoch 12/20
225/225              4s 19ms/step -
accuracy: 0.8686 - loss: 0.4816 - val_accuracy: 0.7991 - val_loss: 1.0896
Epoch 13/20
225/225              5s 16ms/step -
accuracy: 0.8735 - loss: 0.4677 - val_accuracy: 0.8013 - val_loss: 1.1078
Epoch 14/20
225/225              4s 16ms/step -
accuracy: 0.8772 - loss: 0.4566 - val_accuracy: 0.8008 - val_loss: 1.0669
Epoch 15/20
225/225              4s 17ms/step -
accuracy: 0.8792 - loss: 0.4475 - val_accuracy: 0.8036 - val_loss: 1.1018
Epoch 16/20
225/225              4s 17ms/step -
accuracy: 0.8838 - loss: 0.4171 - val_accuracy: 0.7969 - val_loss: 1.1135
Epoch 17/20
225/225              4s 17ms/step -
accuracy: 0.8930 - loss: 0.4003 - val_accuracy: 0.8036 - val_loss: 1.1902
Epoch 18/20
225/225              5s 17ms/step -
accuracy: 0.8898 - loss: 0.4025 - val_accuracy: 0.7980 - val_loss: 1.1940
Epoch 19/20
225/225              4s 17ms/step -
accuracy: 0.8945 - loss: 0.3821 - val_accuracy: 0.8041 - val_loss: 1.2188
Epoch 20/20
225/225              4s 18ms/step -
accuracy: 0.8953 - loss: 0.3852 - val_accuracy: 0.8030 - val_loss: 1.2051
```

[44]:
```
import matplotlib.pyplot as plt
# Muestro gráfica de accuracy y losses
plt.style.use("ggplot")
plt.figure()
plt.plot(np.arange(0, 20), H.history["loss"], label="train_loss")
```

```python
plt.plot(np.arange(0, 20), H.history["val_loss"], label="val_loss")
plt.plot(np.arange(0, 20), H.history["accuracy"], label="train_acc")
plt.plot(np.arange(0, 20), H.history["val_accuracy"], label="val_acc")
plt.title("Training Loss and Accuracy")
plt.xlabel("Epoch #")
plt.ylabel("Loss/Accuracy")
plt.legend()

# ¿Y si desconectamos mayor porcentaje de neuronas?
```

[44]: <matplotlib.legend.Legend at 0x2134c785950>



[45]:
```python
# Evaluando el modelo de predicción con las imágenes de test
print("[INFO]: Evaluando red neuronal...")
model_drop.predict(x_test)
loss, accuracy = model_drop.evaluate(x_test, y_test)
print('Loss {}, accuracy {}'.format(loss,accuracy))
```

```
[INFO]: Evaluando red neuronal…
71/71              0s 5ms/step
```

```
71/71            0s 5ms/step -
accuracy: 0.8010 - loss: 1.3470
Loss 1.346963882446289, accuracy 0.800979495048523
```

- Batch Normalization

```
[46]: from tensorflow.keras.models import Sequential
      from tensorflow.keras.layers import Dense, BatchNormalization
      # Vamos a codificar la topología de nuestro MLP
      model_bn = Sequential()
      model_bn.add(Dense(128,activation='relu', input_shape=(WORD_LIMIT,)))
      model_bn.add(BatchNormalization())
      model_bn.add(Dropout(0.6))
      model_bn.add(Dense(64,activation='relu'))
      model_bn.add(BatchNormalization())
      model_bn.add(Dropout(0.6))
      model_bn.add(Dense(46,activation='softmax')) # Reparto de la unidad de␣
       ↪probabilidad entre num_classes
```

```
E:\anacconda\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarning:
Do not pass an `input_shape`/`input_dim` argument to a layer. When using
Sequential models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
[47]: # Ahora que tengo definida la arquitectura, la compilo
      model_bn.compile(optimizer='adam',
                    loss='categorical_crossentropy', # ideal para clasificacion␣
       ↪multiclase
                    metrics=['accuracy'])
```

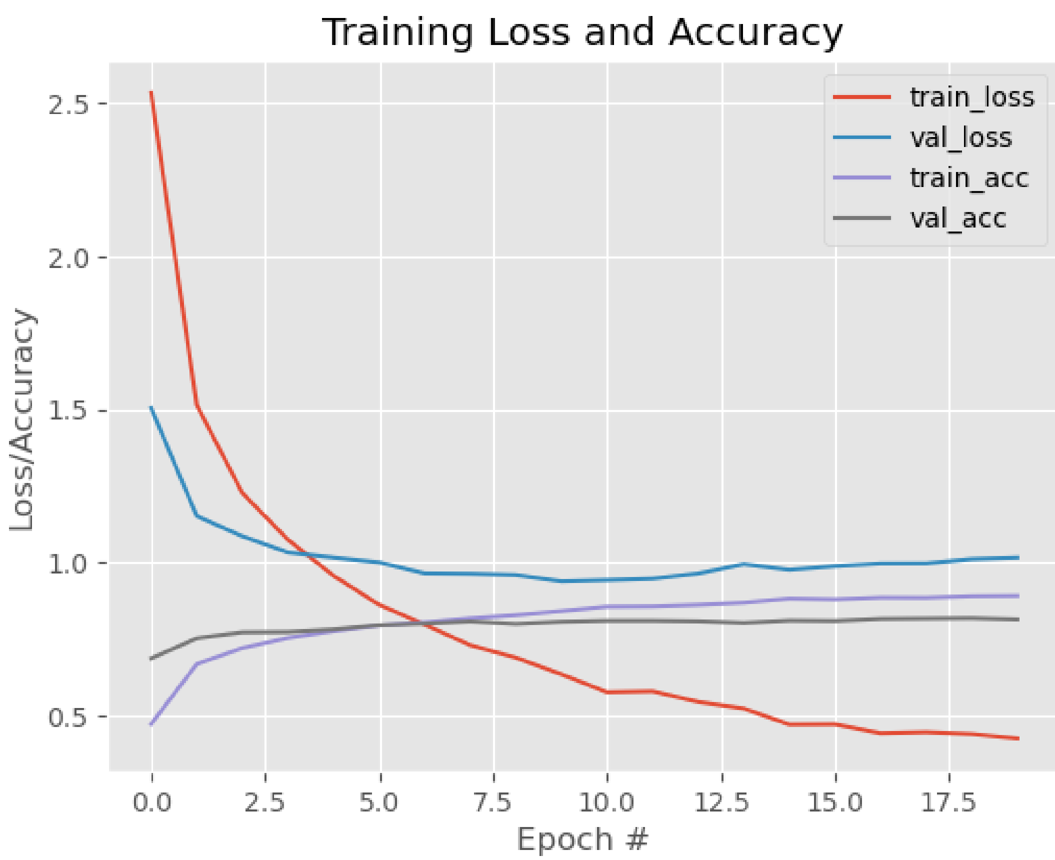
```
[48]: # A entrenar nuestra red neuronal sea dicho!
      H = model_bn.fit(x_train,y_train,epochs=20,batch_size=32, validation_split=0.2)
```

```
Epoch 1/20
225/225          8s 20ms/step -
accuracy: 0.4717 - loss: 2.5336 - val_accuracy: 0.6861 - val_loss: 1.5040
Epoch 2/20
225/225          4s 16ms/step -
accuracy: 0.6676 - loss: 1.5140 - val_accuracy: 0.7518 - val_loss: 1.1513
Epoch 3/20
225/225          4s 18ms/step -
accuracy: 0.7200 - loss: 1.2265 - val_accuracy: 0.7713 - val_loss: 1.0853
Epoch 4/20
225/225          5s 17ms/step -
accuracy: 0.7528 - loss: 1.0742 - val_accuracy: 0.7724 - val_loss: 1.0324
Epoch 5/20
225/225          5s 17ms/step -
accuracy: 0.7755 - loss: 0.9564 - val_accuracy: 0.7807 - val_loss: 1.0156
```

```
Epoch 6/20
225/225             5s 18ms/step -
accuracy: 0.7939 - loss: 0.8616 - val_accuracy: 0.7947 - val_loss: 0.9998
Epoch 7/20
225/225             4s 17ms/step -
accuracy: 0.8036 - loss: 0.7966 - val_accuracy: 0.7997 - val_loss: 0.9641
Epoch 8/20
225/225             5s 17ms/step -
accuracy: 0.8170 - loss: 0.7294 - val_accuracy: 0.8069 - val_loss: 0.9627
Epoch 9/20
225/225             5s 18ms/step -
accuracy: 0.8274 - loss: 0.6885 - val_accuracy: 0.7986 - val_loss: 0.9587
Epoch 10/20
225/225             5s 19ms/step -
accuracy: 0.8406 - loss: 0.6343 - val_accuracy: 0.8058 - val_loss: 0.9382
Epoch 11/20
225/225             4s 19ms/step -
accuracy: 0.8550 - loss: 0.5757 - val_accuracy: 0.8091 - val_loss: 0.9422
Epoch 12/20
225/225             4s 17ms/step -
accuracy: 0.8559 - loss: 0.5784 - val_accuracy: 0.8091 - val_loss: 0.9468
Epoch 13/20
225/225             4s 17ms/step -
accuracy: 0.8611 - loss: 0.5443 - val_accuracy: 0.8075 - val_loss: 0.9635
Epoch 14/20
225/225             6s 22ms/step -
accuracy: 0.8683 - loss: 0.5224 - val_accuracy: 0.8019 - val_loss: 0.9940
Epoch 15/20
225/225             4s 19ms/step -
accuracy: 0.8816 - loss: 0.4703 - val_accuracy: 0.8097 - val_loss: 0.9759
Epoch 16/20
225/225             4s 19ms/step -
accuracy: 0.8791 - loss: 0.4710 - val_accuracy: 0.8086 - val_loss: 0.9876
Epoch 17/20
225/225             4s 16ms/step -
accuracy: 0.8849 - loss: 0.4419 - val_accuracy: 0.8152 - val_loss: 0.9960
Epoch 18/20
225/225             4s 20ms/step -
accuracy: 0.8845 - loss: 0.4453 - val_accuracy: 0.8164 - val_loss: 0.9964
Epoch 19/20
225/225             4s 19ms/step -
accuracy: 0.8895 - loss: 0.4391 - val_accuracy: 0.8175 - val_loss: 1.0106
Epoch 20/20
225/225             5s 17ms/step -
accuracy: 0.8906 - loss: 0.4247 - val_accuracy: 0.8136 - val_loss: 1.0154
```

[49]:
```
import matplotlib.pyplot as plt
# Muestro gráfica de accuracy y losses
plt.style.use("ggplot")
plt.figure()
plt.plot(np.arange(0, 20), H.history["loss"], label="train_loss")
plt.plot(np.arange(0, 20), H.history["val_loss"], label="val_loss")
plt.plot(np.arange(0, 20), H.history["accuracy"], label="train_acc")
plt.plot(np.arange(0, 20), H.history["val_accuracy"], label="val_acc")
plt.title("Training Loss and Accuracy")
plt.xlabel("Epoch #")
plt.ylabel("Loss/Accuracy")
plt.legend()
```

[49]:
```
<matplotlib.legend.Legend at 0x2134cf291d0>
```



[50]:
```
# Evaluando el modelo de predicción con las imágenes de test
print("[INFO]: Evaluando red neuronal...")
model_bn.predict(x_test)
loss, accuracy = model_bn.evaluate(x_test, y_test)
```

```
print('Loss {}, accuracy {}'.format(loss,accuracy))
```

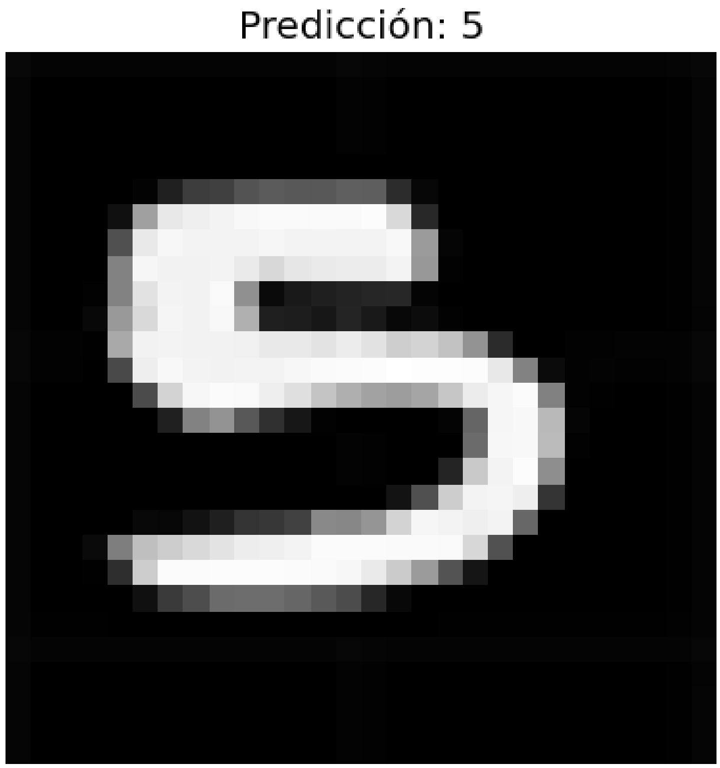
```
[INFO]: Evaluando red neuronal…
71/71             1s 6ms/step
71/71             0s 5ms/step -
accuracy: 0.7952 - loss: 1.1081
Loss 1.1081066131591797, accuracy 0.7951914668083191
```

[51]:
```
print(model.input_shape)
```

```
(None, 10000)
```

[52]:
```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image, ImageOps

# usar el modelo de MNIST guardado antes
modelo_final = model_mnist

img = Image.open("img/img/numero.png").convert("L")
img = ImageOps.invert(img)
img = img.resize((28, 28))

img_array = np.array(img, dtype=np.float32) / 255.0
img_entrada = np.expand_dims(img_array, axis=0)   # shape: (1, 28, 28)

pred = modelo_final(img_entrada, training=False).numpy()
numero_predicho = np.argmax(pred, axis=1)[0]
confianza = float(np.max(pred))

print("Predicción:", numero_predicho)
print("Confianza:", confianza)

plt.imshow(img_array, cmap="gray")
plt.title(f"Predicción: {numero_predicho}")
plt.axis("off")
plt.show()
```

```
Predicción: 5
Confianza: 0.5558651685714722
```



GABRIEL SALGUERO

Repo: https://github.com/reromash1/machines2